

Research paper

Enhancing parameter identification for proton exchange membrane fuel cell using modified manta ray foraging optimization

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ABSTRACT

In this paper, an accurate model of proton exchange membrane fuel cell (PEMFC) for optimal identification of PEMFC parameters has been developed. The optimization methodology is based on the modified version of Manta Ray Foraging Optimization (MMRFO) technique for minimizing the sum of squared errors (SSE) between the Experimentally measured stack voltage and the estimated voltage produced by the optimized model. In the modified methodology, the sine-cosine method has been utilized to enhance the global searching capability in the exploration phase and the local searching capability in the exploitation phase of the MRFO algorithm. In order to validate the effectiveness of the suggested methodology, four different case studies comprising standard benchmark 250 W PEMFC, BCS-500 W PEMFC, SR-12 500 W FC, and 1 kW Temasek stacks were utilized, and the attainments have been compared with the measured polarization characteristics. The attainments have been intensively compared with several metaheuristic algorithms (MA) including Tree growth Algorithm (TGA), Grey wolf optimizer (GWO), Whale optimization algorithm (WOA), Salp swarm algorithm (SSA), and original Manta Ray Foraging Optimization (MRFO), to confirm the superiority of the MMRFO against the compared techniques. The obtained results give a satisfactory agreement between the MMRFO-based model and the experimentally measured data. Finally, the achievements confirmed the effectiveness of the MMRFO over the basic MRFO algorithm and other novel metaheuristic algorithms in identifying PEMFC parameters.

1. Introduction

Hydrogen gas has a number of important characteristics that qualify it to be the future fuel, as it is a pure and environmentally safe fuel. It does not emit harmful gases when burned and has high energy, so it is one of the unique sources of energy as fuel or as an energy carrier in fuel cells (FCs), as it can be used either directly or when mixed with natural gas in specific proportions (Zhang et al., 2016; Edwards et al., 2007). Hydrogen can be used to produce energy in several ways, including using it to run internal combustion engines for cars and vehicles, or in fuel cells to produce electric current. Hydrogen is also used as fuel in spacecraft and rockets (Edwards et al., 2007). Hydrogen is considered a

renewable energy resource and can be obtained through a large number of sources and in different ways, such as the electrolysis of water, which breaks the chemical bonds between oxygen and hydrogen in the water. However, this method is financially costly because it consumes large amounts of electrical energy (Yamamura et al., 2022). Therefore, the process of electrolysis of water is carried out at high temperatures to reduce the amount of energy required for the analysis process (Becherif et al., 2015). Fuel cells are called the source of energy for the twenty-first century, and fuel cell (FC) is a device that converts chemical energy stored in hydrocarbon compounds to electrical energy. The fuel used in these cells is hydrogen, natural gas, or methanol in the presence of oxygen or atmospheric air (Lucia, 2014).

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Fuel cells (FCs) are distinguished from batteries in their reliance on the merging of hydrogen and oxygen (obtained from external sources) to generate electric energy (Lucia, 2014; Kordesch and Simader, 1996). This gives FCs interest compared to batteries, while in traditional batteries the internal elements are the basis for power production, where the chemical reaction of the components of the battery takes place to generate electricity, and the reaction holds on until the end of the interacting chemicals, so the conventional battery stops until it is recharged again. While FCs operate continuously because their fuels come from outward sources (Kordesch and Simader, 1996).

There are different types of FCs, and these cells are classified based on the chemicals they use, as well as the electrolyte plates. The most common and popular type is the PEMFC. This electrochemical cell is used to produce electrical energy by continuously supplying the cell with oxygen and hydrogen gases (Kordesch and Simader, 1996; Carrette et al., 2000). At anode side, hydrogen is oxidized into protons that circulate through the electrolyte to cathode side, and electrons that circulate from outside the cell to cathode, where everyone meets oxygen, which is reduced to form water (Carrette et al., 2000). There are several sorts of FCs that can be classified according to the type of electrolyte that determines the end use of the fuel cell. For example, those that use a conductive polymer for protons are the most widely used in electric cars, while those that use ceramics that conduct oxygen ions are better for home use to supply electricity and heat. Finally, FCs are one of the most environmentally friendly means of generating energy. That is why it receives great attention for its development and harnessing it in the future in driving cars and supplying homes with electric current. This kind of fuel is friend to the environment and a good alternative to the current fuel used in transportation, in addition to its great importance for poor countries or those suffering from a lack of traditional energy resources (Carrette et al., 2000; Vielstich et al., 2003). PEMFCs are among the best kinds of FCs for use as an electric source for transportation that will be used in the future as an alternative to internal combustion engines (ICE) that use diesel as fuel. The first use of these cells was in the sixties by NASA in the (NASA Gemini program) (Peighambardoust et al., 2010). These cells use a solid polymers membrane (thin plastic film) as a medium. This polymer is permeable to protons saturated with water and does not pass electrons. The fuel used in PEM cells is hydrogen and the charge carriers are positive hydrogen ions (protons). At the positive electrode, hydrogen splits into hydrogen ions (protons) and electrons. Hydrogen ions cross the medium to the negative electrode, while electrons flow through an external circuit to generate electricity. Oxygen (from the air) merges with electrons and

hydrogen ions, forming water (Peighambardoust et al., 2010; Devanathan, 2008).

Compared to other types of FC, PEMFCs produce more energy for a given cell weight or volume. This high-power density means they are both lightweight and highly reliable. Additionally, PEMFCs operate at temperatures below 100°C, allowing them to start quickly once operational conditions are met. Their rapid response to fluctuating electricity demands positions them as a leading candidate for various electrical applications. Furthermore, the solid-state nature of these cells offers advantages over liquid-based systems. Constructing the cells becomes simpler, particularly when connecting the anode and cathode, which can lead to reduced manufacturing costs. The solid-state also offers increased resistance to challenges like corrosion, enhancing the cell's lifespan. This durability is perhaps the most significant advantage of solid-state systems, ensuring the longevity of the fuel cell and, by extension, the entire stack (Wee, 2007).

In general, modeling of PEMFC is a substantial issue that can be utilized for optimal designing of the FC stack and exploring the electrochemical characteristics of the fuel cell (Fei et al., 2019; Yuan et al., 2020). This modeling depends on physical and electrochemical procedures to regulate the efficiency of FCs. The mathematical model comprises a differential equation system to realize the transference actions and their correspondences. Based on the discussed cases and because of the prominence of mathematical modeling of PEMFCs, numerous research papers were conducted on this topic (Cao et al., 2019; Eslami et al., 2019; Sultan et al., 2023; Menesy et al., 2021a).

In recent years, metaheuristic algorithms have gained significant attention in diverse engineering applications, particularly in the optimization of real-world problems. In the context of optimizing parameter extraction in PEM fuel cell modeling, understanding the development and utilization of various algorithms is pivotal. For instance, a comprehensive study compared the efficiency of the political optimization algorithm, the Archimedes optimization algorithm, and the Levy flight algorithm, particularly in design optimization relevant to the industry (Yildiz et al., 2021a). Similarly, a conceptual comparison among algorithms such as the ecogeographic-based algorithm, equilibrium algorithm, marine predators algorithm, and slime mold algorithm has been made with a focus on optimal product design (Yildiz et al., 2021a). This array of algorithms offers a robust foundation for potential applications in parameter extraction, suggesting the vast adaptability and flexibility of such techniques. The slime mold algorithm, combined with the kriging surrogate model, has been shown to enhance the crash-worthiness of electric vehicles, suggesting its efficacy in complex

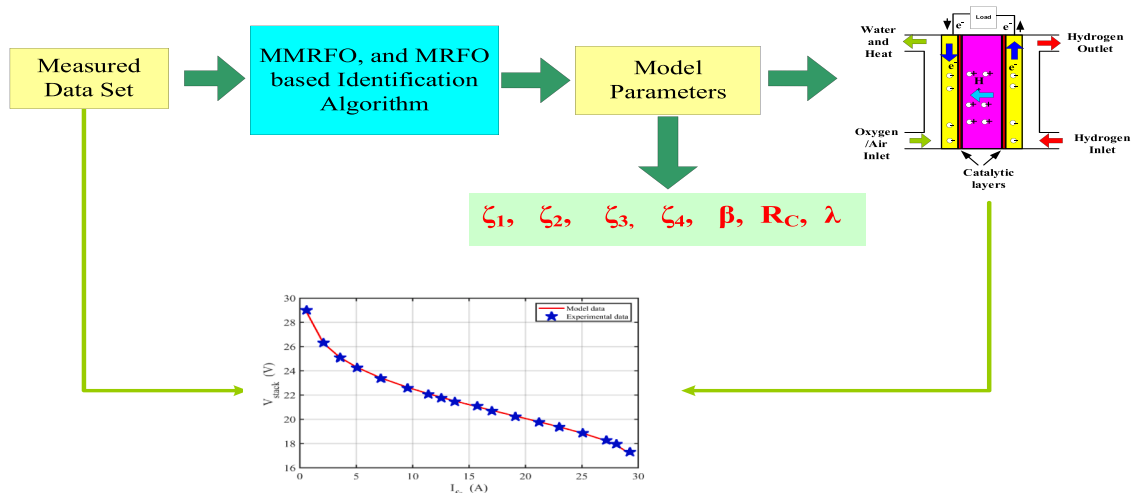


Fig. 1. Graphical representation of the optimization process for PEMFC parameter estimation and validation of the MMRFO.

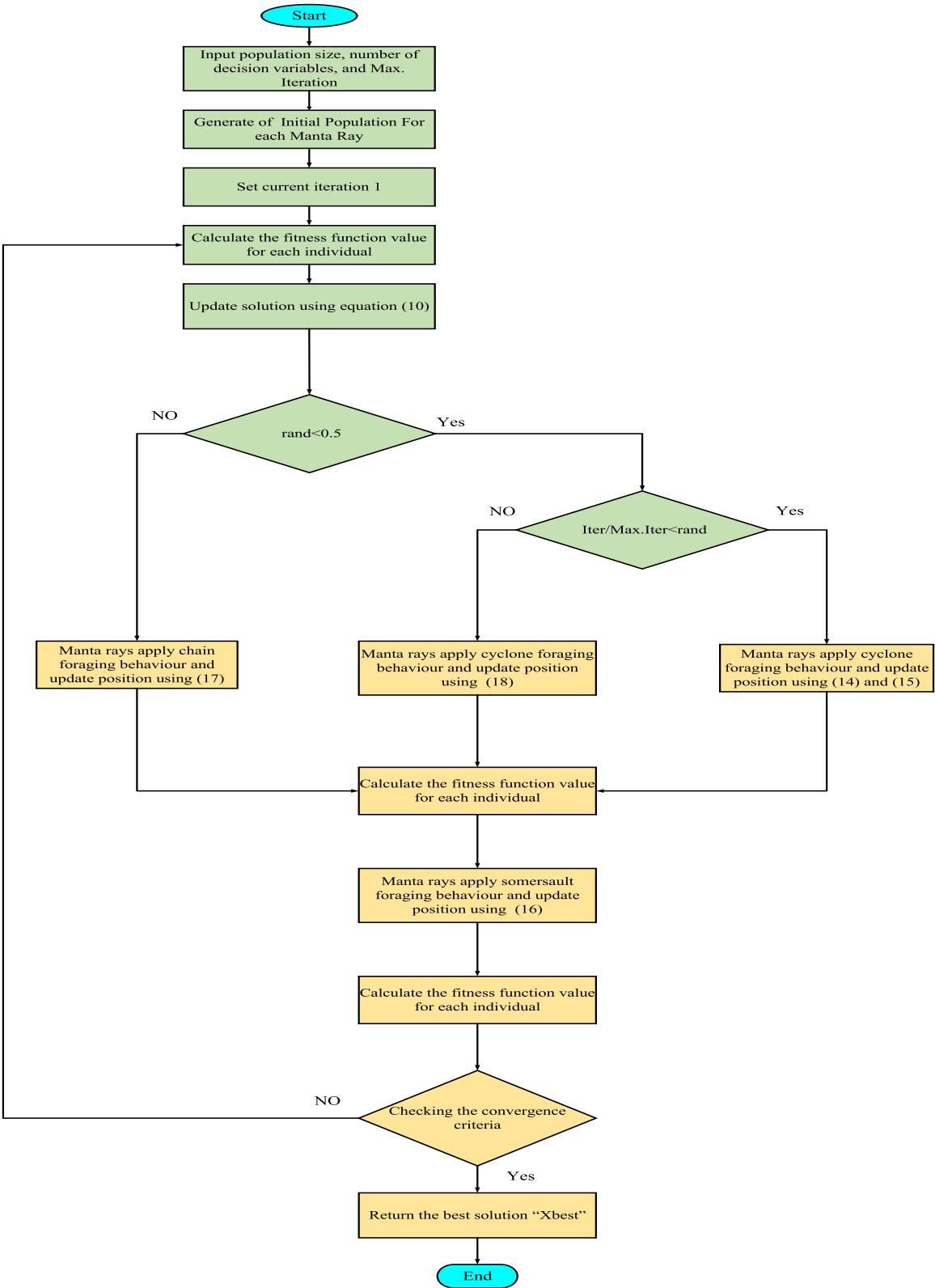


Fig. 2. Flowchart of the MMRFO.

modeling scenarios (Yıldız, 2020a). This mirrors other hybrid techniques such as the salp swarm algorithm combined with the radial basis function used in electric vehicle component design (Yıldız, 2020b). The realm of vehicle crashworthiness optimization further extends with the employment of multi-surrogate-assisted metaheuristics (Aye et al., 2019). Algorithms like the enhanced grasshopper optimization with

elite opposition-based learning have been leveraged for real-world engineering problems, showcasing their potential in diverse applications (Yıldız et al., 2022). The Henry gas solubility optimization algorithm's incorporation of chaos theory similarly suggests its adaptability in complex engineering situations (Yıldız et al., 2021b). Furthermore, the grasshopper optimization algorithm's adaptation for the robust design

Table 1
Design parameters and operating conditions of the studied PEMFCs.

FC type	<i>N</i> (cells)	<i>A</i> (cm ²)	<i>l</i> (μm)	<i>J</i> _{max} (mA/cm ²)	<i>P</i> H ₂ (atm)	<i>P</i> _{O₂} (atm)	<i>T</i> (K)
250 W stack	24	27	127	860	1.00	1.00	343.15
BCS 500 W	32	64	178	469	1.00	0.2095	333
SR–12 PEM 500 W	48	62.5	25	672	1.47628	0.2095	323
Temasek 1 kW	20	150	51	1500	0.50	0.50	323

of robot grippers signifies its broad-spectrum utility (Yildiz et al., 2021). The advent of the chaotic Lévy flight distribution optimization algorithm, tailored for constrained engineering problems, further echoes the growing versatility and prominence of such algorithms in optimization tasks (Yildiz et al., 2022). Given the breadth of these developments, the focus on a Modified Manta Ray Foraging Approach in the context of PEM fuel cell modeling is timely and stands to benefit from insights drawn from these diverse algorithmic advancements.

Inverse modeling in PEMFCs focuses on identifying model parameters or operational states based on observable data, such as current and voltage, to improve the understanding and performance of FCs. This approach is pivotal for optimizing FC design, enhancing efficiency, and ensuring long-term stability. The use of root means square error (RMSE) as a metric for evaluating the accuracy of these models is common, as it provides a clear measure of the difference between observed and predicted values, thereby offering insights into the model's predictive performance. For literature on inverse PEMFC modeling, the works by Pukrushpan et al (Pukrushpan et al., 2004). and Amphlett et al (Baumert, 1994)., are foundational. Pukrushpan et al (Pukrushpan et al., 2004). discuss the dynamic modeling of fuel cell operations, focusing on the control and simulation of PEMFC systems. Although not directly inverse modeling work, it sets a significant groundwork for understanding fuel cell dynamics, which is crucial for developing and evaluating inverse models. Amphlett et al (Baumert, 1994). provide one of the early models for predicting the performance of PEMFCs based on empirical data. Their work, focusing on the relationship between the operational parameters and the fuel cell's performance, offers a basis for subsequent inverse modeling efforts, where the goal is to deduce operational states from performance data.

The authors in (Ćalasan et al., 2023), introduce a novel approach to

parameter estimation in proton exchange membrane fuel cells (PEMFCs) through the application of the iterative Lambert W function, significantly diverging from traditional models by expressing PEMFC current as a function of voltage. This new mathematical modeling aims to enhance the accuracy in simulating PEMFC behavior, and is backed by optimization algorithms including a modified evaporation rate water cycle algorithm (ChERWCA). The results demonstrate a marginal improvement in the sum-of-squares error (SSE) for voltage, reducing by approximately 0.1 % compared to the best previously reported results. However, the model's complexity and computational demands may restrict its practical utility. Validation of the model is limited to two specific PEMFC configurations, which could impact its generalizability across different types or operational conditions. Additionally, the model's effectiveness heavily relies on the performance of complex optimization algorithms, which could influence its robustness in various settings. Rahman et al (Rahman et al., 2019)., investigated mathematical model as well as experimental results for a PEMFC operation under open circuit voltage and restricted current conditions. The obtained results have been evaluated by restricted current tests and polarization characteristics based on various operating scenarios. Results presented that the suggested 1-D dry model gives proper agreement with measured attainments. Bao et al. proposed a new procedure for optimal identification of PEMFC parameters using an improved Monarch Butterfly Optimization (IMBO) algorithm (Bao et al., 2020). Two PEMFCs, namely 6 kW NedSstack PS6 and 2 kW Nexa FC were used in this study. The obtained results confirmed the good agreement between the results obtained from IMBO and the measured ones. Duan et al., Proposed a new modification for the crow search algorithm (CSA) and applied the modified version for estimating the parameters of several PEMFCs. The total of square errors between the experimental and estimated voltages

Table 2
The search space of the optimized parameters for validation.

Parameter	ξ_1	$\xi_2 \times 10^{-3}$	$\xi_3 \times 10^{-5}$	$\xi_4 \times 10^{-5}$	λ (V)	R_c (mΩ)	β
upper	−0.8532	6.00	9.8	−0.954	23.00	0.80	0.50
lower	−1.1997	0.80	3.6	−2.60	10.00	0.10	0.0136

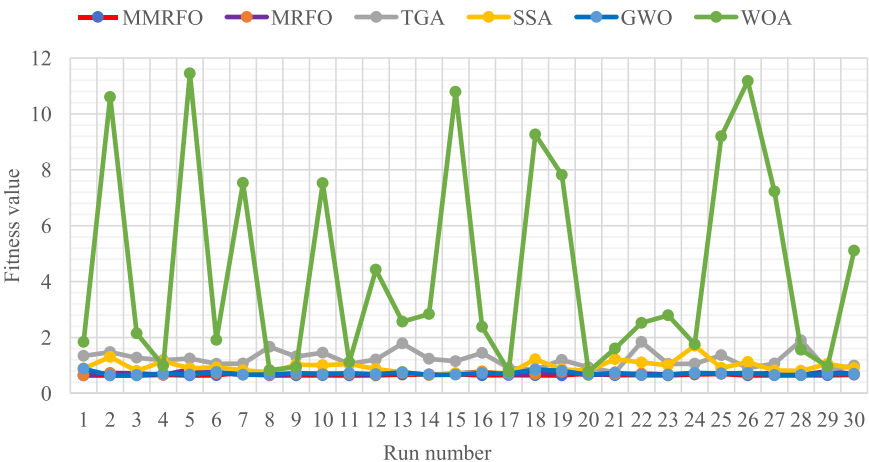


Fig. 3. The end values of the objective function (SSE) of all algorithms for 250 W PEMFC.

Table 3
Comparison of the achievements of the MMRFO with other algorithms for 250 W PEMFC.

	MMRFO	MRFO	TGA	SSA	GWO	WOA
1	0.642466758	0.645217942	1.340351172	0.889377894	0.878835714	1.839355687
2	0.642388369	0.728887244	1.475503715	1.303023966	0.64440856	10.60327218
3	0.642554959	0.716786789	1.274668377	0.798235207	0.654867615	2.147706167
4	0.662176981	0.659621048	1.186602047	1.151252637	0.703446178	0.991045414
5	0.642446608	0.861058015	1.244233593	0.892703651	0.667378241	11.44906733
6	0.646765276	0.721704277	1.057975809	0.929414055	0.756680877	1.912902533
7	0.757570877	0.671433769	1.064269273	0.825803939	0.665718559	7.532101178
8	0.642332179	0.655188211	1.664234747	0.753992965	0.67309835	0.831004028
9	0.644621664	0.692350639	1.318541632	1.028171436	0.726974934	0.941415909
10	0.646903866	0.706624719	1.453949038	1.006120018	0.678692455	7.522103326
11	0.642587901	0.688116339	1.052608725	1.055148786	0.721237866	1.113179824
12	0.643951973	0.665842529	1.208337794	0.869027349	0.686726862	4.424758365
13	0.669833813	0.737148246	1.786024834	0.756412148	0.753223391	2.565925732
14	0.672099379	0.664729788	1.232677071	0.650047239	0.659339303	2.835376999
15	0.686776279	0.700876135	1.151882523	0.718052147	0.668069934	10.78934735
16	0.643463645	0.779739192	1.444916605	0.746147643	0.728258875	2.378050239
17	0.653314329	0.675654103	0.908109765	0.716769049	0.699978406	0.790318749
18	0.647147108	0.690395373	0.853089108	1.21694211	0.85407312	9.261243898
19	0.642082942	0.766234033	1.196329663	0.854373211	0.8057173	7.817557434
20	0.712802158	0.660838806	0.930564755	0.814305574	0.664061032	0.779270025
21	0.647241457	0.683650014	0.749606946	1.221398468	0.732578129	1.603046869
22	0.657103152	0.707024593	1.839883288	1.106279063	0.653002648	2.52304535
23	0.642924812	0.677118134	1.052159975	1.006410269	0.667725088	2.79115446
24	0.67421921	0.716774643	1.05910084	1.703295758	0.728105474	1.756048427
25	0.694381374	0.708930035	1.362580834	0.908814904	0.71232621	9.198620905
26	0.642665009	0.690294138	0.903296751	1.123252975	0.723100656	11.17717955
27	0.646600837	0.731555194	1.066268426	0.816707746	0.657708713	7.226124848
28	0.647985777	0.662842173	1.893472882	0.805281775	0.65442218	1.558439984
29	0.646631856	0.789822977	0.841203425	1.054317038	0.691983653	0.964806679
30	0.664151395	0.692482336	0.988280048	0.90023018	0.695897146	5.106361049
Best	0.642082942	0.645217942	0.749606946	0.650047239	0.796926618	0.796926618
Worst	0.757570877	0.861058015	1.893472882	1.703295758	0.924600759	0.924600759
Mean	0.658273065	0.704964714	1.220024122	0.95404364	0.829441969	0.829441969
Median	0.646834571	0.692416488	1.191465855	0.896466915	0.816662317	0.816662317
SD	2.582516663	4.662354611	29.7779324	22.00535548	3.165014838	3.165014838
RE	0.756450062	2.777981	18.8265535	14.02958351	1.224028021	1.224028021
MAE	0.016190123	0.059746773	0.470417176	0.303996401	0.03251535	0.03251535
RMES	0.030113585	0.075305866	0.554084014	0.373126332	0.045006542	0.045006542
Eff.	97.67400988	91.88190771	64.86418508	71.15782326	96.20719293	96.20719293

is incorporated as the objective function (Duan et al., 2022). Hachana et al (Hachana and El-Fergany, 2022)., developed artificial bee colony differential evolution optimizer (ABCDE) and to validate the proposed algorithm they apply it for solving the problem of parameter extraction of PEMFC, while five commercial fuel cell stacks have been tested. Rezk et al (Rezk et al., 2022a)., proposed the Gradient-based Optimizer (GBO) for identifying the best parameters of PEMFCs. The SSE is utilized as the objective function. The obtained results proved the superiority of the GBO over recent optimization algorithms in extracting the optimal parameters of PEMFC. Ashraf et al (Ashraf et al., 2022)., utilized honey badger optimizer (HBO) for identifying the parameters of Ballard Mark, SR-12 and 250 W PEMFC stacks. Moreover, a sensitivity analysis is conducted to illustrate the impact of change in the values of the optimized parameters. Rezk et al (Rezk et al., 2022b)., proposed novel optimization approach utilizing bald eagle search algorithm (BES) for extracting the parameters of PEMFCs. BCS 500 W and NedStack PS6 PEMFC stacks are used to verify the effectiveness of the the BES algorithm. A satisfying matching between the measured and estimated polarization characteristic has been confirmed. Wilberforce et al (Wilberforce et al., 2023)., proposed five optimization techniques namely, Grey Wolf Optimization(GWO), Particle Swarm Optimization (PSO), Slime Mould Algorithm (SMA), Harris Hawk Optimizer (HHO), and artificial ecosystem-based algorithm(AEO) for extracting the seven unknown parameters of PEMFCs. The AEO algorithm presented the best results over other optimization techniques. Chen et al (Chen and Zhang, 2022)., proposed Improved African vulture optimizer (IAVO) for parameters identification of NedStackPS6, BCS 500 W, and SR-12 500 W fuel cell stacks.

Based on the aforementioned literature review, it can be summarized that the application of metaheuristic optimization techniques to give better performance in modeling PEMFCs is intensively utilized. The key

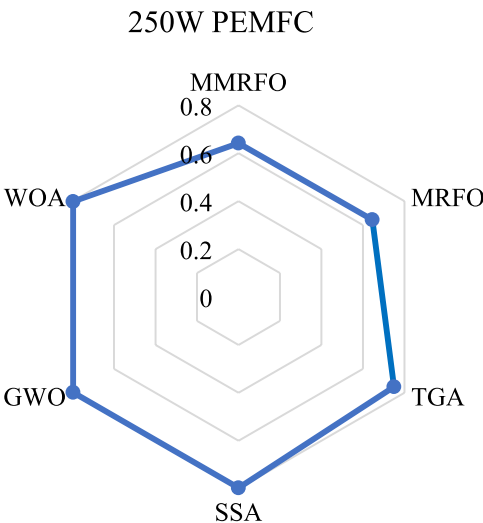


Fig. 4. Radar plot for the best OF obtained so far from all algorithms for 250 W PEMFC.

contributions and novelties of this study are presented as follows:

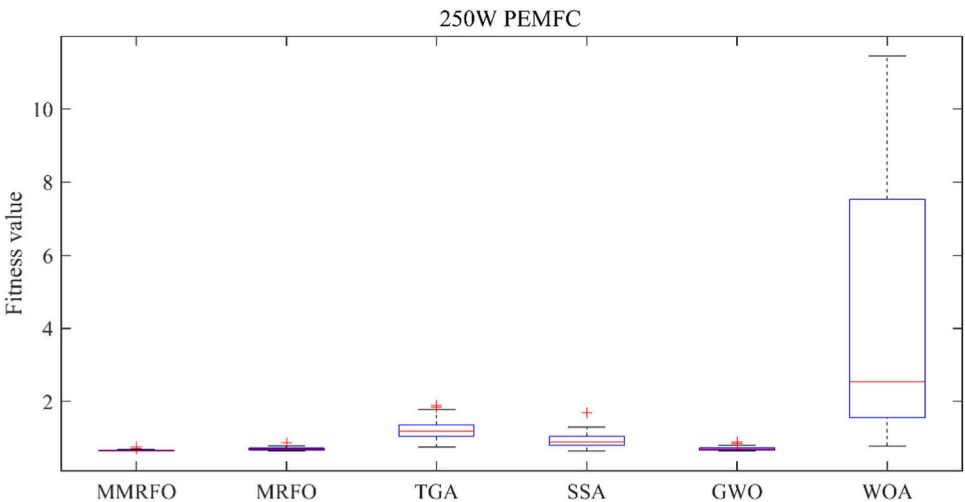


Fig. 5. Box-plot for 250 W PEMFC.

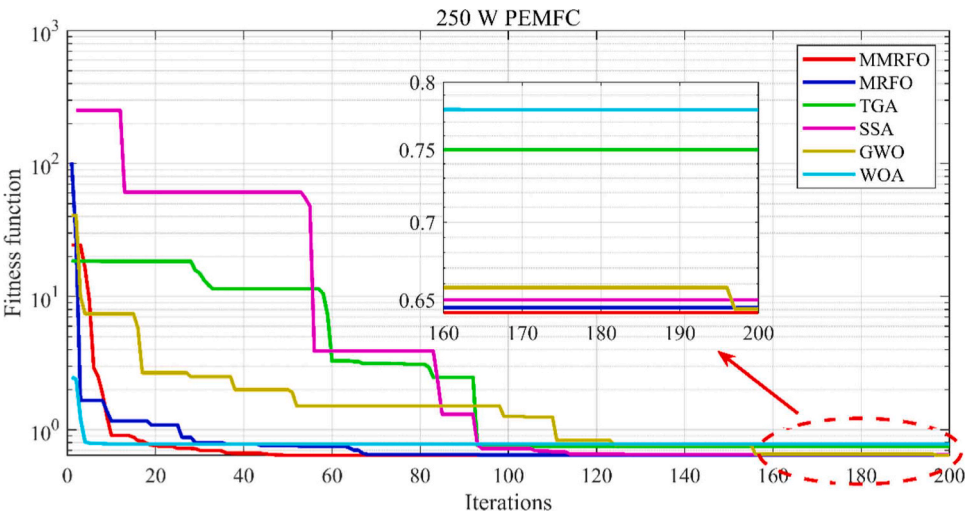
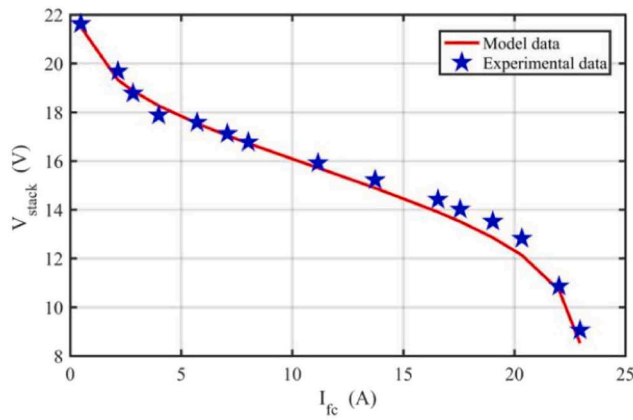


Fig. 6. Convergences curves of the best run for all algorithms.

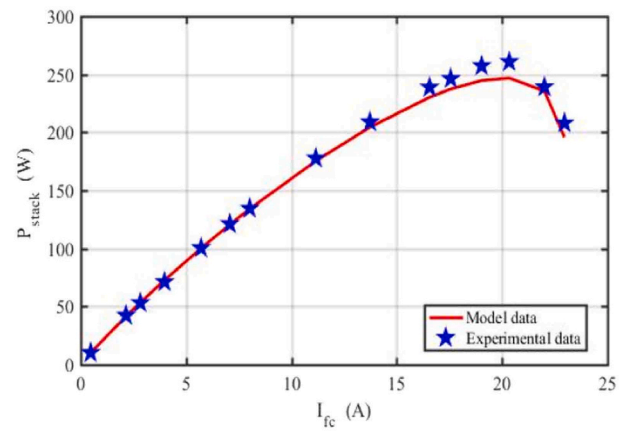
Table 4
Extracted parameters of 250 W PEMFC based on the MMRFO and other algorithms.

	ξ_1	$\xi_2 \times 10^{-3}$	$\xi_3 \times 10^{-5}$	$\xi_4 \times 10^{-4}$	λ	b	$R_c \times 10^{-4}$	SSE
MMRFO	−1.06414	3.244598	6.73197	−1.56013	18.36909	0.05452	2.26552	0.64208
MRFO	−1.11417	2.85814	7.74401	−1.46370	21.93193	0.05038	2.14608	0.64522
TGA	−1.1914	4.112	6.057	−1.709	18.68	0.0544	4.852	0.7496
SSA	−1.1896	3.16068	0.3997	−1.5604	23	0.054210	0.0001	0.65004
GWO	−0.9745	2.8913	0.6185	−1.550	22.99230	0.05472	0.0001	0.6444
WOA	−0.9200	2.6990	0.6003	−1.437	19.13034	0.05376	0.0001	0.7793
HHO (Shaheen et al., 2021)	−0.89794	2.578	5.13	−11.303	14.24547	0.014029	0.506	1.3815
VSDE (Fathy et al., 2020)	−1.1921	3.199	3.799	−1.870	22.81	0.0290	1.202	1.0526
MVO (Fathy and Rezk, 2018)	−0.9182	3.129	8.703	−1.802	15.19	0.0180	4.223	3.5846
SBO (Duan et al., 2019)	−1.0554	3.795	9.800	−1.175	23.00	0.0136	1.088	1.1510
ASO (Agwa et al., 2019)	−1.1132	3.600	9.800	−2.000	22.17	0.0248	1.000	0.7346
CHHO4 (Menesy et al., 2019)	−0.91017	2.96613	7.78812	−1.52389	22.2369	0.053665	1.93212	0.674734

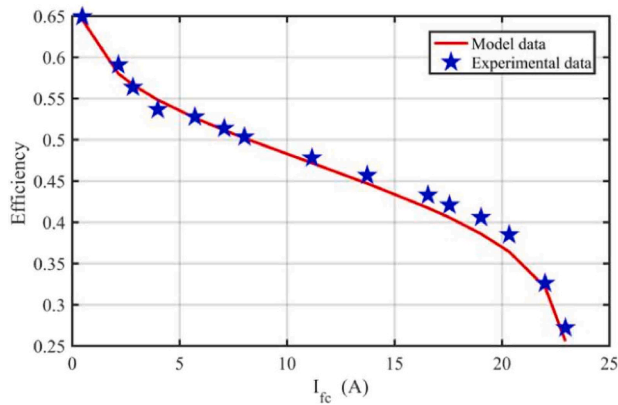
- PEMFC Parameter Optimization: Applying the MMRFO algorithm for precise identification and optimization of PEMFC parameters.
- Comparative Algorithm Analysis: Benchmarking MMRFO against other metaheuristic algorithms like TGA, SSA, GWO, and WOA, highlighting its relative strengths.
- Mathematical Model Validation: Rigorous validation of the MMRFO-based PEMFC model through comparison with standard polarization curves.
- Impact Study on Operational Conditions: Investigation of the effects of varying operational conditions, such as cell temperature and reactant pressure, on PEMFC performance.
- Statistical Measurement Validation: Providing conclusive evidence and approval for the superiority of the MMRFO technique over conventional methods through robust statistical measurement analysis.



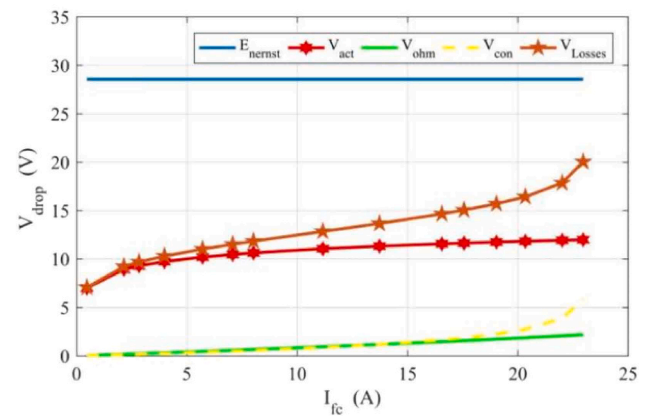
(a)



(b)



(c)



(d)

Fig. 7. Characteristics of 250 W PEMFC: (a) V/I curves, (b) P/I curves, (c) efficiency, (d) voltage losses.

- Sensitivity Analysis: Detailed analysis of the impact of changes in optimized parameters on the objective function, enhancing the understanding of parameter sensitivities in PEMFCs.

The rest of the paper is arranged as follow: In section two, PEMFC mathematical model is presented. The third section presents the formulation of the optimization problem. The fourth section presents the original MRFO algorithm and the MMRFO algorithm. Simulation results, statistical analysis, and sensitivity study are presented in the fifth section. The sixth section presents the conclusion and main outcomes of the paper.

2. Methodology

2.1. PEMFC Model

The voltage/current profile of a PEMFC exhibits nonlinearity due to various voltage drops within the FC. In theoretical terms, the open circuit voltage of an individual cell spans from 0.9 to 1.23 V, contingent upon specific operational parameters such as cell temperature and gas pressure. Consequently, multiple cells, denoted as n_c are serially aligned to augment the voltage output from the PEMFC stack. According to Kirchhoff's voltage law, PEMFC stack voltage can be expressed as follows (Amphlett et al., 1996; Sultan et al., 2021a; Menesy et al., 2021b),

$$V_{fc} = n_c \times (E_{Nernst} - \eta_{act} - \eta_{\Omega} - \eta_{con}) \quad (1)$$

where, η_{act} implies the activation over-voltage that presents the delayed reactions from the activation barrier; η_{Ω} represents the ohmic over-voltage that represents the variations in the FC resistance associated the conductivity of the electrolyte, and η_{con} presents the concentration over-voltage that results from the mass transfer processes. E_{Nernst} represents the voltage when the circuit is open, caused by thermodynamic balance or Nernst voltage that is formulated as follows (Menesy et al., 2020a; Abd El Monem et al., 2014),

$$E_{Nernst} = 1.229 - 0.85 \times 10^{-3} (T_{fc} - 298.15) + 4.3085 \times 10^{-5} \times T_{fc} \times \left(\ln(P_{H_2}) + \frac{\ln(P_{O_2})}{2} \right) \quad (2)$$

where, P_{H_2} signifies the anode side's hydrogen partial pressure in (atm.), and P_{O_2} indicates the cathode side's oxygen partial pressure in (atm.), while T_{fc} symbolizes the operating temperature of the PEMFC (K). The activation overvoltage η_{act} can be calculated as follow (Yu et al., 2019):

$$\eta_{act} = \xi_1 + \xi_2 T + \xi_3 T_{fc} \ln \left(\frac{P_{O_2}}{5.08 \times 10^6 \times \exp(-498/T_{fc})} \right) + \xi_4 T_{fc} \ln(I_{fc}) \quad (3)$$

where, I_{fc} stands for FC current; ξ_1 , ξ_2 , ξ_3 , and ξ_4 imply the empirical parameters that have to be optimally identified (Rao et al., 2019; Sultan et al., 2020).

Ohmic overvoltage arises from the resistance to electron movement in the collector plates and electrodes, as well as the resistance to proton

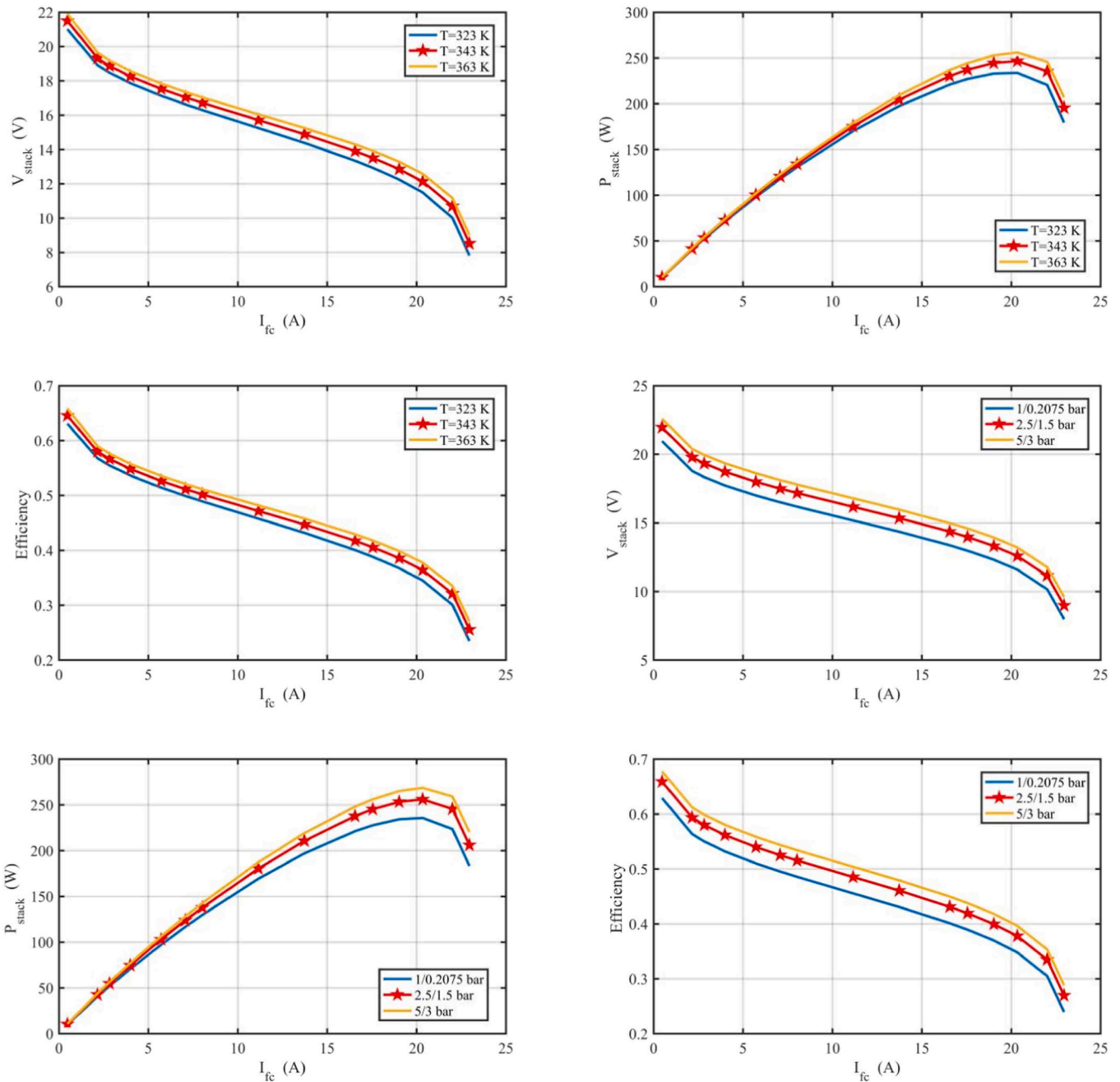


Fig. 8. Operation of 250 W PEMFC under different temperature and fuel pressure.

movement in the membrane. It is represented as follows (Hachana and El-Fergany, 2022):

$$\eta_{\Omega} = I_{fc}(\Omega_m + \Omega_c) \quad (4)$$

where, Ω_c implies to the concentration resistance; Ω_m is the ohmic membrane resistance that is calculated as follow (Menesy et al., 2021a; Amphlett et al., 2022):

$$\Omega_m = \rho_m \left(\frac{l}{A_m} \right) \quad (5)$$

where, l implies the depth of the polymer membrane (μm); A_m symbolizes the membrane area (cm^2); ρ_m denotes the material's specific resistivity from which the membrane is produced ($\Omega\cdot\text{cm}$). By the way, the Nafion membrane resistivity can be calculated as given in Eq. (6) (Menesy et al., 2021b, 2020a):

$$\rho_m = \frac{181.6 \left[1 + 0.03 \left(\frac{I_{fc}}{A_m} \right) + 0.062 \left(\frac{T_{fc}}{303} \right)^2 \left(\frac{I_{fc}}{A_m} \right)^{2.5} \right]}{\left[\lambda - 0.0634 - 3 \left(\frac{I_{fc}}{A_m} \right) \right] \times e^{\left[4.18 \left(1 - \frac{303}{T_{fc}} \right) \right]}} \quad (6)$$

where, λ is a coefficient influenced by the membrane preparation method and needs to be determined. The concentration over voltage that depends on the dimensions of the FC and in particular the active area A_m , can be expressed as follow (Baumert, 1994; Rezk et al., 2022a),

$$\eta_{con} = -\beta \ln \left(1 - \frac{I_{fc}}{A_m J_{max}} \right) \quad (7)$$

where, J_{max} symbolizes the maximum current density of PEMFC (A/

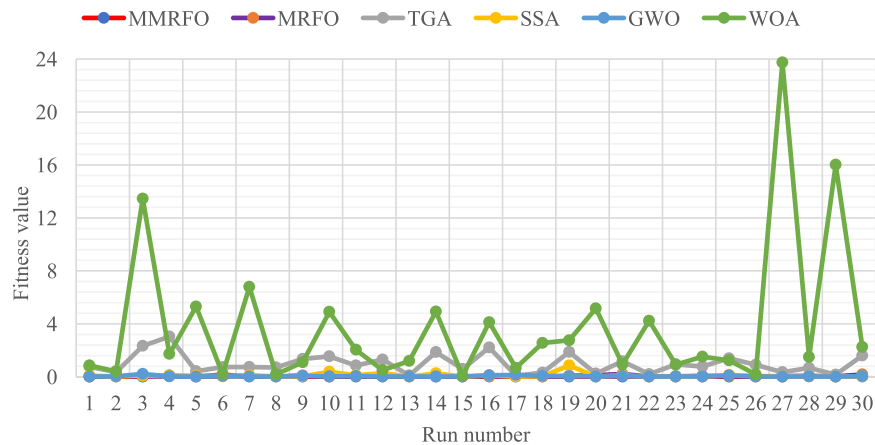


Fig. 9. The end values of the objective function (SSE) of all algorithms for BCS 500 W PEMFC.

Table 5
Comparison of the achievements of MMRFO with other algorithms for BCS 500 W PEMFC.

Algorithm	MMRFO	MRFO	TGA	SSA	GWO	WOA
1	0.012474583	0.020559599	0.753910045	0.08557596	0.019907663	0.859613267
2	0.016484299	0.029434103	0.356854501	0.018870708	0.036762268	0.409227222
3	0.012035475	0.046902676	2.332373872	0.095954952	0.216794428	13.45712019
4	0.093687273	0.038719119	3.038015387	0.099966098	0.024405096	1.722806865
5	0.012255397	0.036449497	0.418926169	0.089160254	0.020283063	5.307958428
6	0.168013285	0.026281232	0.737089305	0.060992451	0.060624049	0.199725081
7	0.014555871	0.070600659	0.739798053	0.094045138	0.021135411	6.789511284
8	0.030107601	0.017990447	0.70211703	0.031579217	0.022774259	0.180373271
9	0.011574603	0.094948304	1.350956111	0.031523151	0.068280197	1.095665966
10	0.011594199	0.01558216	1.540304912	0.382595488	0.034162415	4.906003576
11	0.011731345	0.055192593	0.856567414	0.118077781	0.052504547	2.041280112
12	0.024110376	0.02278264	1.30026467	0.277760619	0.02081613	0.498923473
13	0.026816133	0.01717182	0.09919333	0.013397587	0.025231238	1.191763636
14	0.011614347	0.024888169	1.86102157	0.261676373	0.025244739	4.925485836
15	0.022733743	0.024352691	0.589421799	0.020342834	0.018536093	0.078228431
16	0.014331874	0.023613179	2.215636785	0.101536137	0.112633529	4.119640206
17	0.011589757	0.041829997	0.083524999	0.023054238	0.13886353	0.668584727
18	0.028167656	0.012951386	0.312269748	0.032233377	0.063895089	2.55664868
19	0.01250098	0.068177922	1.87741823	0.859302483	0.04750227	2.752148602
20	0.011694172	0.116725868	0.243152713	0.022459003	0.030231608	5.166978928
21	0.012891489	0.212664535	1.169873712	0.110238067	0.053448361	0.936759293
22	0.024397523	0.013605415	0.197222558	0.017639438	0.017317919	4.232102567
23	0.011750149	0.028406566	0.938723886	0.023829958	0.027617672	0.916704404
24	0.019215552	0.069621539	0.773347458	0.017927245	0.018419737	1.520758279
25	0.011709297	0.042193724	1.394088242	0.138144407	0.098637316	1.237026939
26	0.01463081	0.022361269	0.90411996	0.068151046	0.027015744	0.180714772
27	0.011685324	0.052039153	0.346706788	0.025486056	0.01267661	23.73912938
28	0.043354836	0.07218305	0.67394684	0.017182138	0.027284866	1.500846408
29	0.022497142	0.02841888	0.163007887	0.015190102	0.012742367	16.01620083
30	0.017994446	0.175184351	1.604863243	0.028732837	0.049425204	2.254949912
Best	0.011574603	0.012951386	0.083524999	0.796926618	0.01267661	0.078228431
Worst	0.168013285	0.212664535	3.038015387	0.924600759	0.216794428	23.73912938
Mean	0.024939985	0.050727751	0.985823907	0.829441969	0.046839114	3.715429352
Median	0.014443873	0.0329418	0.763628752	0.816662317	0.027451269	1.621782572
SD	3.131417776	4.659308578	73.88976703	3.165014838	4.420390755	528.3956432
RE	34.64148667	87.50345272	324.0822228	1.224028021	80.84772551	1394.838495
MAE	0.013365382	0.037776366	0.902298908	0.03251535	0.034162504	3.637200921
RMES	0.033563749	0.05937681	1.158410158	0.045006542	0.05528046	6.341825677
Eff.	71.99333194	42.22368206	19.08595011	96.20719293	43.88014235	11.96353358

cm^2); β is the diffusion empirical coefficient that has to be optimally determined.

2.2. Problem Formulation

From the PEMFC mathematical model outlined in the preceding subsection, one can observe that the model incorporates a predefined set of empirical parameters, denoted as, $X = [\xi_1, \xi_2, \xi_3, \xi_4, \lambda, \Omega_c, \beta]$. Identifying the parameter fuel cells can be formulated as a constrained

optimization challenge. In this problem, the objective function (OF) measures the accurateness of MMRFO-based PEMFC model while reducing the deviations between attainments of the model based on the identified parameters and measured polarization curves reported in the manual of FC stack. PEMFC generated voltage and experimentally measured voltage are compared, and the SSE is taken as the OF (Li et al., 2010; Menesy et al., 2020b),

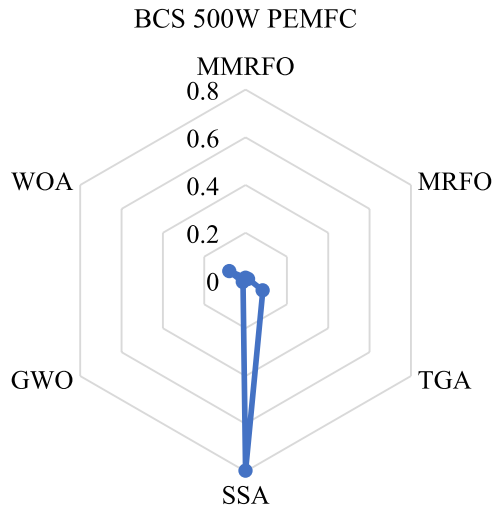


Fig. 10. Radar plot for the best OF obtained so far from all algorithms for BCS 500 W PEMFC.

$$SSE = \sum_{i=1}^N (V_{s,meas} - V_{s,model})^2 \quad (8)$$

where, N symbolizes the span of measured data points, $V_{s,meas}$ is the experimentally determined fuel cell voltage, and $V_{s,model}$ symbolizes the voltage obtained based on the optimized model. In addition, a set of

inequity constraints (IC) is applied on the objective function (SSE) as given in Eq. (9) (Sultan et al., 2020; Menesy et al., 2020b).

$$IC = \begin{cases} \xi_{n,min} \leq \xi_n \leq \xi_{n,max} \forall n \in 1 : 4 \\ \lambda_{min} \leq \lambda \leq \lambda_{max} \\ \Omega_{c,min} \leq \Omega_c \leq \Omega_{c,max} \\ \beta_{min} \leq \beta \leq \beta_{max} \end{cases} \quad (9)$$

A brief description of the optimization process of the PEMFC parameter estimation and validation of the provided MMRFO is graphically provided in Fig. 1.

3. Optimization algorithms

3.1. MRFO

Manta ray foraging optimization (MRFO), a new metaheuristic method, mimics the foraging behaviors of manta rays. Manta rays may travel in a group or alone, and they are often observed foraging in groups. Three manta ray foraging strategies serve as the inspiration for MRFO, including chain foraging, cyclone foraging, and somersault foraging. MRFO algorithm begins by initiating a random population in the search space. Each individual updates its location with respect to both the one in front of it and the reference position in each iteration. Manta rays line up behind one another in an organized line when 50 or more of them begin foraging. Smaller male manta rays swim on top of female ones and piggyback on their backs in time with the females' pectoral fin beats (Zhao et al., 2020; Yildiz and Mehta, 2022).

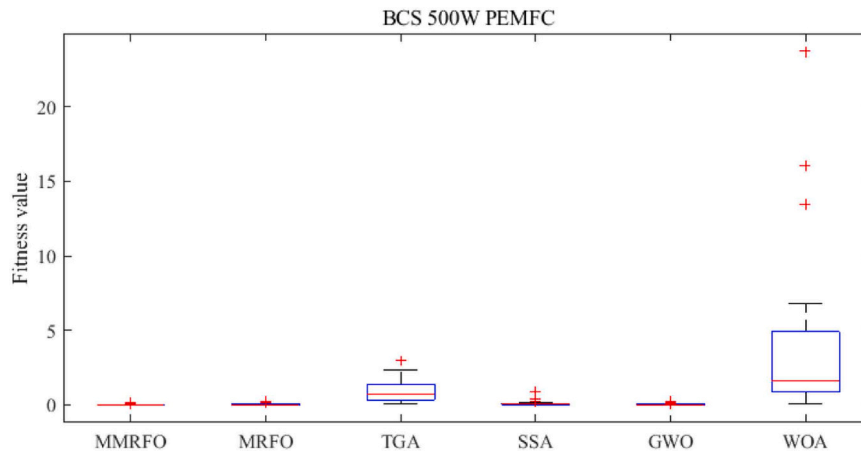


Fig. 11. Box-plot for BCS 500 W PEMFC.

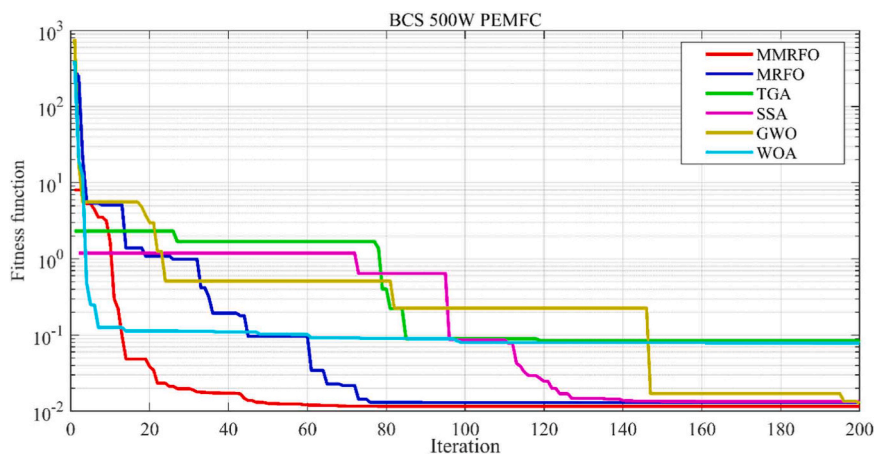
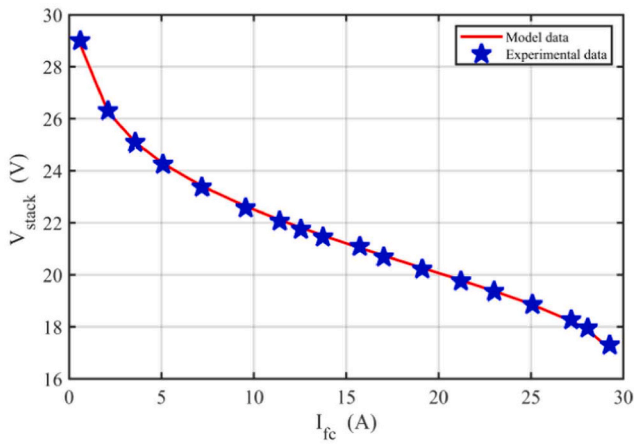


Fig. 12. Convergence curves of the best run for all algorithms.

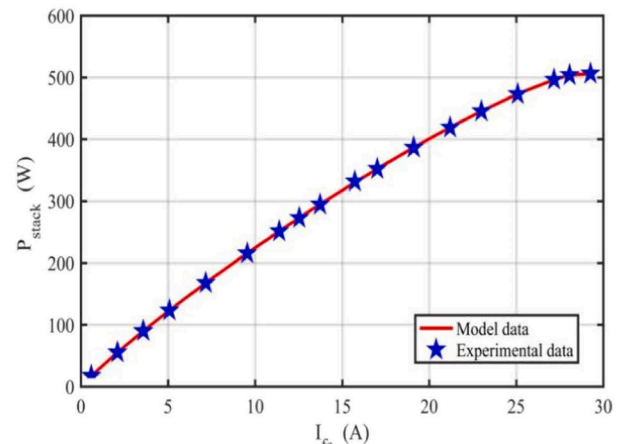
Table 6

Extracted parameters of BCS 500 W PEMFC based on the MMRFO and other methods.

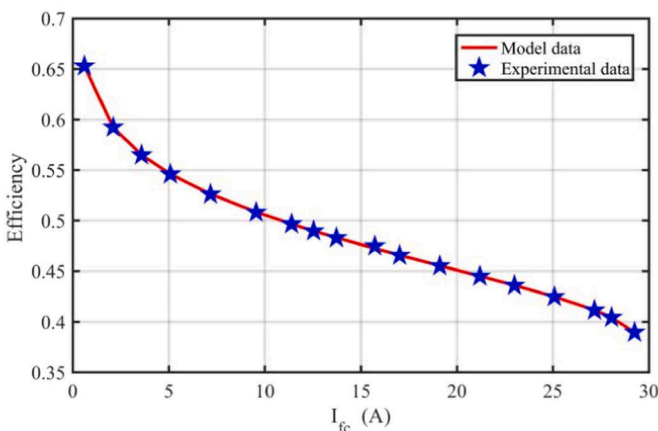
	ξ_1	$\xi_2 \times 10^{-3}$	$\xi_3 \times 10^{-5}$	$\xi_4 \times 10^{-4}$	λ	β	$R_c \times 10^{-4}$	SSE
MMRFO	−1.14206	3.14415	4.85346	−1.92980	21.071183	0.0161669	1.162537	0.0115746
MRFO	−1.08700	5.88945	7.56310	−1.07567	14.49902	0.17965	5.08060	0.01295
TGA	−0.9704	2.952	5.952	−1.838	22.502	0.0136	3.831	0.08352
SSA	−1.018	2.3151	5.240	−1.2815	18.8547	0.0136	7.5036	7.1889
GWO	−1.1365	2.9254	3.7688	−1.3949	18.5446	0.013600	8.00	5.5604
WOA	−0.9797	5.3559	5.743	−1.8017	16.4712	0.016317	0.1	0.078228
EBES (Alsaïdan et al., 2022)	−1.1997	3.3559	5.44	−19.3017	20.8772	0.016126	0.1	0.0116977
BES (Alsaïdan et al., 2022)	−0.8532	0.002446	6.37	−19.3	20.8772	0.016126	0.1	0.0116977
MOA (Shaheen et al., 2021)	−1.117899	3.718476	9.8	−19.10714	22.99758	0.0158144	0.39573	0.012900
GWO-CS (Shaheen et al., 2021)	−1.04401	3.251	6.82	−19.118	22.48551	0.015624	0.362	0.012853
HHO (Sultan et al., 2021a)	−0.88072	2.227	3.89	−12.3163	14.0556	0.027168	0.268	3.7088
NNA (Fawzi et al., 2019)	−1.0596	3.7435	9.6902	−19.302	20.8772	0.0161	0.1	0.011698
ICA (Kandidayeni et al., 2019)	−0.9657	2.4798	4.4583	−19.3	22.6626	0.0162	0.246	0.011856
GHO (El-Fergany, 2018)	−1.018	2.3151	5.24	−12.815	18.8547	0.0136	0.7504	7.1889
CHHO4 (Menesy et al., 2019)	−1.16596	3.909533	8.108802	−1.80468	19.34238	0.01360168	6.490139	0.059288



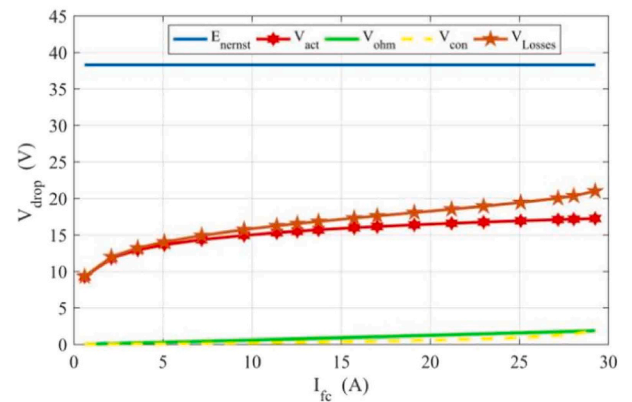
(a)



(b)



(c)



(d)

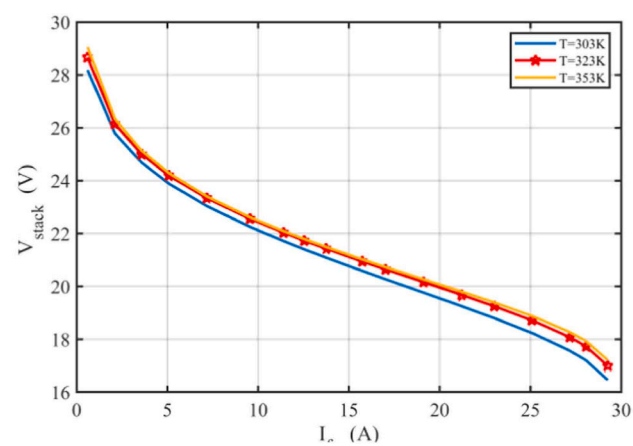
Fig. 13. Characteristics of BCS 500 W PEMFC: (a) V/I curves, (b) P/I curves, (c) efficiency, (d) voltage losses.

3.2. Chain foraging

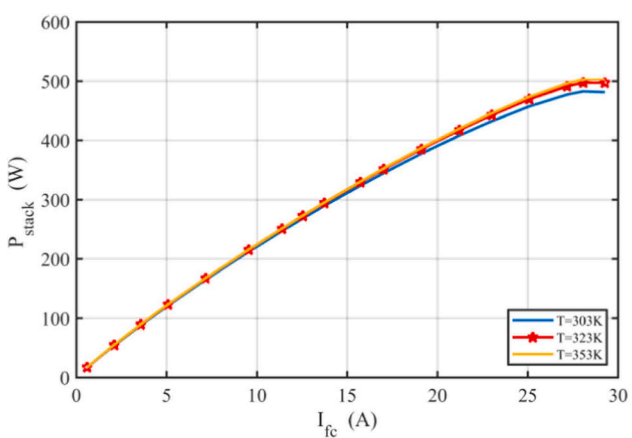
In MRFO, manta rays have the ability to detect plankton and swim toward them. The better site where the bigger the concentration of plankton there is. Manta rays form a foraging chain by aligning themselves from head to tail. In each cycle, each ray updates itself based on the best solution found up to that point and the solution presented by the

ray ahead of it. Below is the depiction of this chain foraging mathematical model (Zhao et al., 2020):

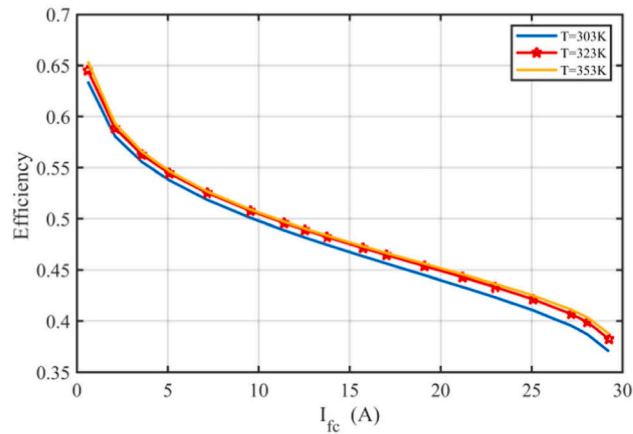
$$x_m^n(i+1) = \begin{cases} x_m^n(i) + rnd \cdot (x_{best}^n(i) - x_m^n(i)) + A \cdot (x_{best}^n(i) - x_m^n(i)) & m = 1 \\ x_m^n(i) + rnd \cdot (x_{m-1}^n(i) - x_m^n(i)) + A \cdot (x_{best}^n(i) - x_m^n(i)) & m = 2, \dots, k \end{cases} \quad (10)$$



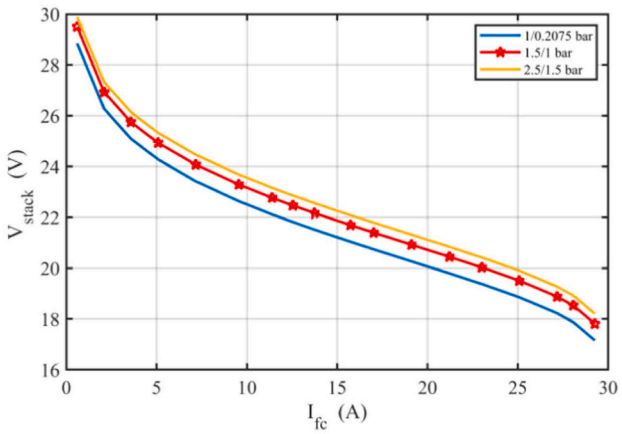
(a)



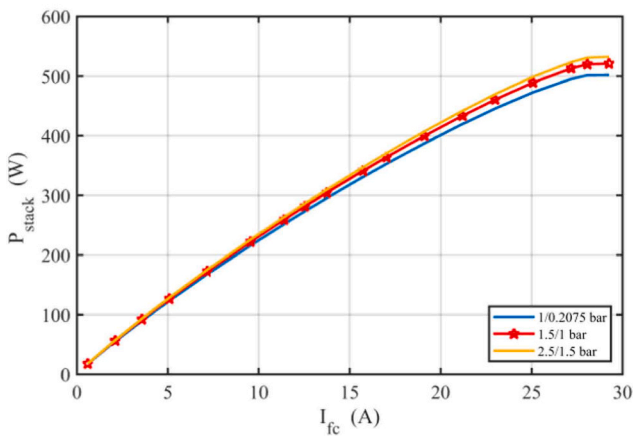
(b)



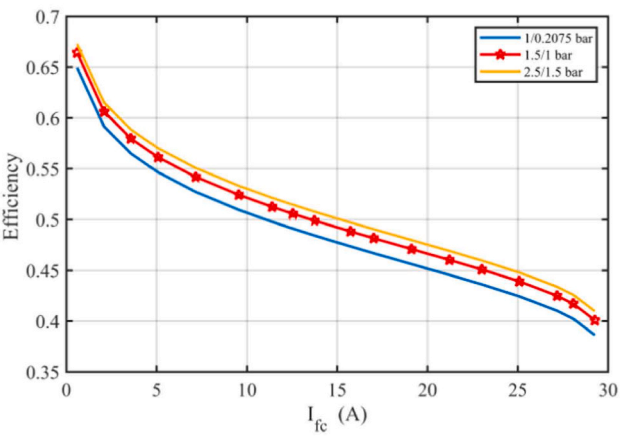
(c)



(d)



(e)



(f)

Fig. 14. Operation of BCS 500 W PEMFC under different temperature and fuel pressure.

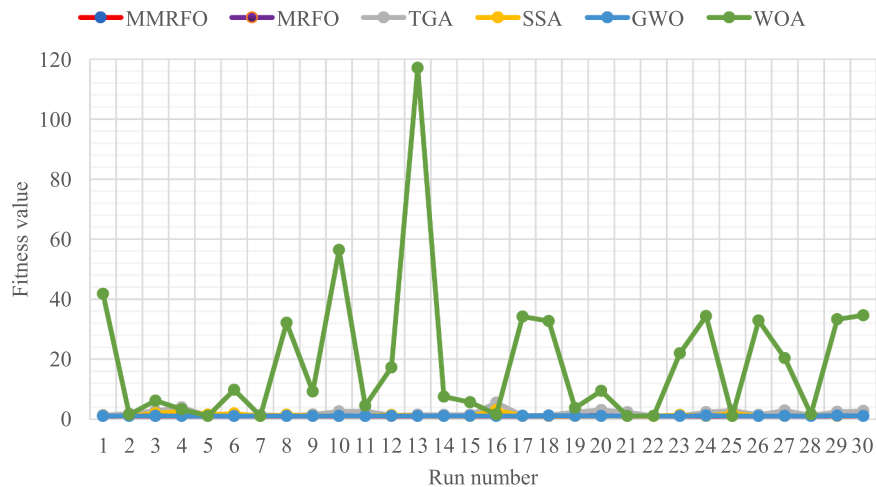


Fig. 15. The end values of the convergence curves of all methods for SR-12 500 W PEMFC.

Table 7
Comparison of the achievements of MMRFO with other algorithms for SR-12 500 W PEMFC.

Algorithm	MMRFO	MRFO	TGA	SSA	GWO	WOA
1	1.061635492	1.069418996	1.333391975	1.15340603	1.070461848	41.7642671
2	1.060494809	1.115136457	1.850064118	1.06124696	1.079290883	1.556401857
3	1.061116655	1.287521733	2.765684512	1.967103403	1.142137355	6.175453006
4	1.057831898	1.05925837	3.941924859	2.645226769	1.088693201	3.314397485
5	1.060278345	1.067178971	1.166354435	1.527405816	1.1165885	1.068142794
6	1.057614955	1.064372883	1.432528563	1.908595989	1.066169481	9.808373453
7	1.060392935	1.12116677	1.453320903	1.060300654	1.096713321	1.090947123
8	1.06270216	1.120847012	1.259351645	1.479475587	1.07990852	32.12613687
9	1.059453371	1.101937817	1.572849821	1.173258555	1.064183076	9.243885256
10	1.056737511	1.212152684	2.590071735	1.107409957	1.114492679	56.40574447
11	1.060384066	1.361002285	2.700416789	1.061174755	1.071751045	4.521359656
12	1.063667036	1.075321836	1.307750995	1.388759469	1.116129483	17.27052477
13	1.056651541	1.086134452	1.490125475	1.089530494	1.07315953	117.0822449
14	1.05666682	1.153677571	1.452896893	1.058903869	1.070274041	7.502187734
15	1.064572149	1.263387181	1.489436856	1.072999195	1.105708276	5.679309493
16	1.058656859	1.121667399	5.504109009	2.985387636	1.077821447	1.623712195
17	1.064144704	1.066752748	1.20524302	1.108606203	1.091788933	34.21581491
18	1.057472899	1.069024227	1.300424355	1.07792277	1.173019199	32.7083934
19	1.058588775	1.261704823	2.400649794	1.077987354	1.119342289	3.752329776
20	1.060475445	1.174595748	3.054337236	1.064738288	1.078216548	9.443799023
21	1.059175054	1.079764838	2.319253928	1.063666287	1.073911655	1.069646346
22	1.06230433	1.077113873	1.104085191	1.076901058	1.100395936	1.130801775
23	1.058223524	1.104526662	1.143379429	1.390732065	1.064574637	21.95591252
24	1.061977428	1.102299604	2.348837249	1.11016277	1.218300518	34.38474685
25	1.058040909	1.071829824	2.908633981	1.94931775	1.083116332	1.060529218
26	1.060285712	1.07249952	1.46344499	1.081031391	1.064915286	32.88694318
27	1.06183679	1.067396403	2.883669135	1.06417105	1.103541926	20.35461591
28	1.056965392	1.067754589	1.245228415	1.183396868	1.082541677	1.653148721
29	1.06510872	1.072763329	2.47802018	1.085871789	1.189707809	33.31129687
30	1.065499511	1.072627829	2.740188061	1.077651592	1.074637436	34.63666601
Min	1.056651541	1.05925837	1.104085191	1.058903869	1.064183076	1.060529218
Max	1.065499511	1.361002285	5.504109009	2.985387636	1.218300518	117.0822449
Mean	1.060298527	1.121361215	2.063522452	1.338411412	1.098383096	19.29325776
Median	1.060334889	1.082949645	1.531487648	1.098470226	1.082829004	9.34384214
SD	0.263051561	7.855610597	99.23923214	48.73490581	3.821118292	2418.635069
RE	0.103543652	1.758858257	26.06965299	7.918779565	0.964120362	515.763118
MAE	0.003646985	0.062102844	0.95943726	0.279507544	0.034200019	18.23272854
RMES	0.004470957	0.099106624	1.368405748	0.554722094	0.050804193	29.96519032
Eff.	99.65663406	94.86759967	63.44063221	85.76076683	96.99370847	29.93195211

$$A = 2 \cdot rnd \cdot \sqrt{|\log(rnd)|}$$

(11)

$x_{best}^n(i)$ is the plankton with high concentration.

Where $x_m^n(i)$ is the position of m individual at the time i in n dimension, A is a weight coefficient, and rnd is a random number between (0, 1).

3.3. Cyclone foraging

Manta rays form a long foraging chain and swim towards the food by

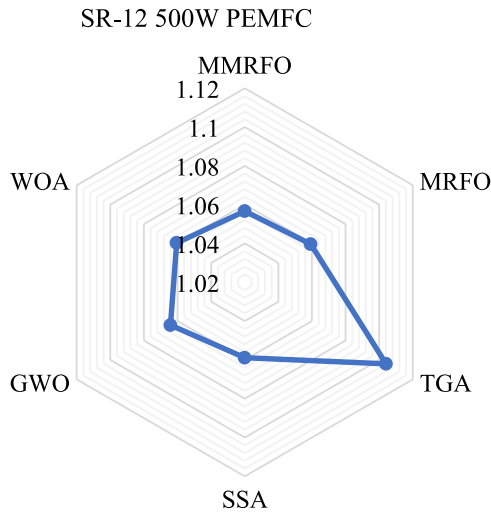


Fig. 16. Radar plot for the best OF obtained so far from all algorithms for SR-12 500 W PEMFC.

a spiral when they identify a patch of plankton position. Each manta ray swims towards the one in front of it in addition to spirally moving towards the plankton. The mathematical model of cyclone foraging can be represented as follows (Zhao et al., 2020):

$$x_m^n(i+1) = \begin{cases} x_{best}^n + rand \cdot (x_{best}^n(i) - x_m^n(i)) + B \cdot (x_{best}^n(i) - x_m^n(i)) & m = 1 \\ x_{best}^n + rand \cdot (x_{m-1}^n(i) - x_m^n(i)) + B \cdot (x_{best}^n(i) - x_m^n(i)) & m = 2, \dots, k \end{cases} \quad (12)$$

$$B = 2e^{rand1 \frac{Max.Iter - Iter + 1}{Max.Iter}} \cdot \sin(2\pi r) \quad (13)$$

where $Max.Iter$ is the maximum number of iterations, $Iter$ is the current iteration, and $rand1$ is the random number between (0, 1). All individuals conduct their searches at random, using the food as their point of reference. MRFO forces each individual to search for a new location, which is a new position far from the current best one by transferring to a new random location in the entire search space. This mechanism can be described as follows (Zhao et al., 2020):

$$x_{rnd}^n = LoB^n + r \cdot (UpB^n - LoB^n) \quad (14)$$

$$x_m^n(i+1) = \begin{cases} x_{rnd}^n + r \cdot (x_{rnd}^n(i) - x_m^n(i)) + B \cdot (x_{rnd}^n(i) - x_m^n(i)) & m = 1 \\ x_{rnd}^n + r \cdot (x_{m-1}^n(i) - x_m^n(i)) + B \cdot (x_{rnd}^n(i) - x_m^n(i)) & m = 2, \dots, k \end{cases} \quad (15)$$

Where, UpB and LoB are the upper and lower boundary limits for the optimization problem, respectively. x_{rnd}^n is a random location that randomly generates in search space.

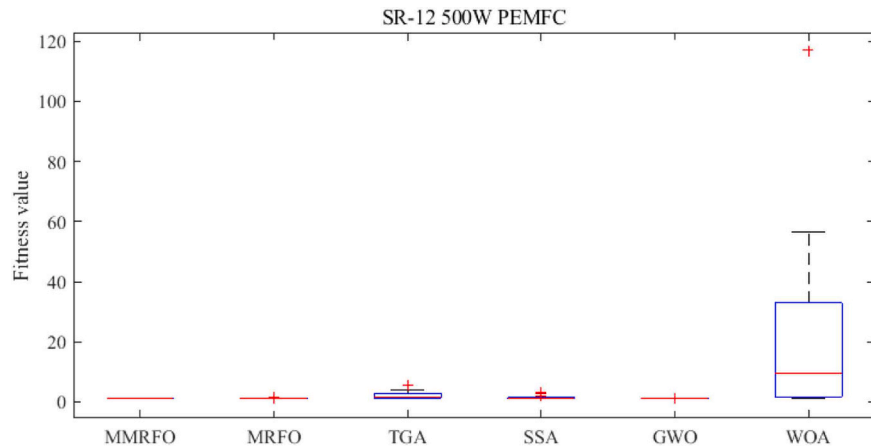


Fig. 17. Box-plot for SR-12 500 W PEMFC.

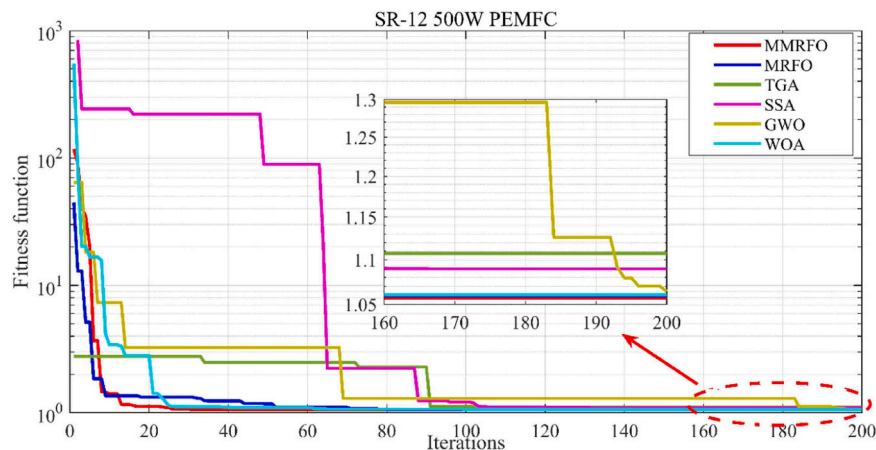
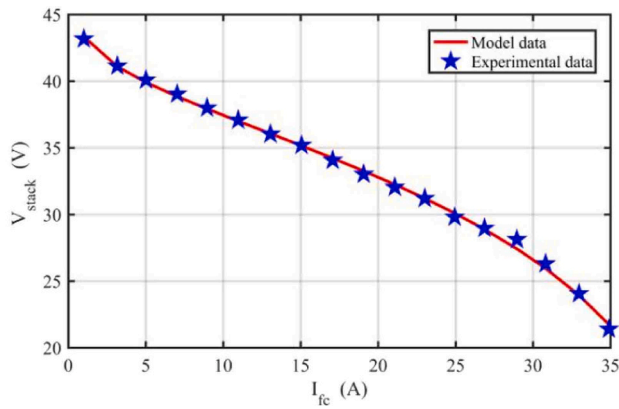


Fig. 18. Convergence curves of the best run for all algorithms for SR-12 500 W.

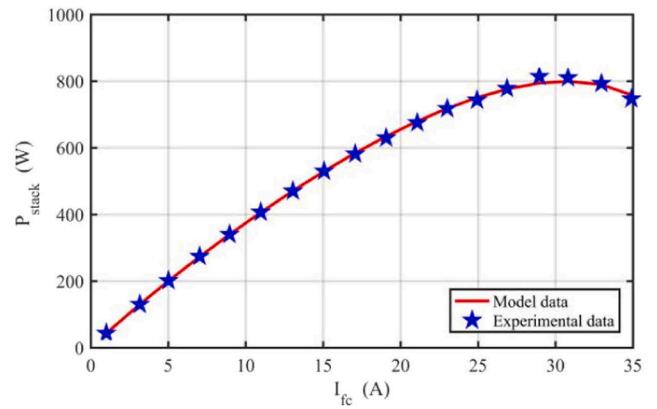
Table 8

Extracted parameters of SR-12 500 W PEMFC based on the MMRFO and other methods.

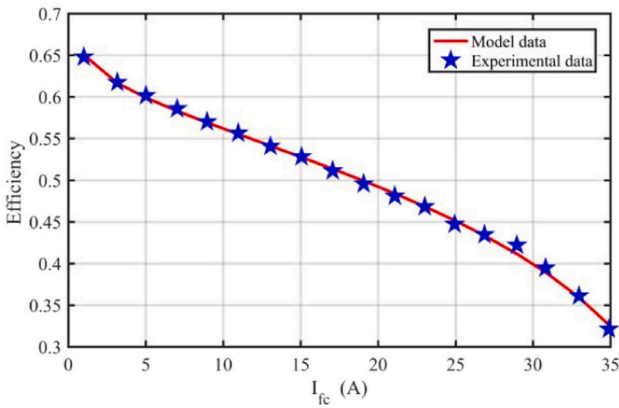
	ξ_1	$\xi_2 \times 10^{-3}$	$\xi_3 \times 10^{-5}$	$\xi_4 \times 10^{-4}$	λ	β	$R_c \times 10^{-4}$	SSE
MMRFO	−1.19141	3.81702	6.325072	−0.95405	21.107816	0.175221	6.76129	1.05665
MRFO	−1.04319	4.73663	6.66458	−1.06193	14.04295	0.477393	4.74369	1.05926
TGA	−1.1123	3.854	4.369	−0.964	23.000	0.1831	2.188	1.1040
SSA	−0.8984	2.4526	0.3603	−0.95400	21.4456	0.1730	0.0007	1.059897
GWO	−1.1763	3.9009	0.7096	−0.9549	22.0990	0.1749	0.0007	1.05970
WOA	−0.8532	3.3563	0.9799	−0.95400	22.3712	0.1776	0.0006	1.06663
HHO (Menesy et al., 2019)	−0.8533	2.417	4.248	−0.954	15.345	0.1779	3.716	1.0593
VSDE (Fathy et al., 2020)	−0.8576	3.010	7.780	−0.954	23.000	0.1516	1.339	1.2660
SBO (Abd El Monem et al., 2014)	−0.9598	2.290	3.230	−0.953	15.863	0.1812	6.742	1.4960
CHHO4 (Menesy et al., 2019)	−0.85320	3.09184	8.23877	−0.9540	22.9115	0.176237	6.24684	1.05716
SSO (Rao et al., 2019)	−0.9664	2.2833	3.400	−0.954	15.7969	0.1804	6.6853	1.517
CS-EO (Chen and Wang, 2019)	−1.0353	3.3540	7.2428	−0.954	10	0.1471	7.1233	7.5753



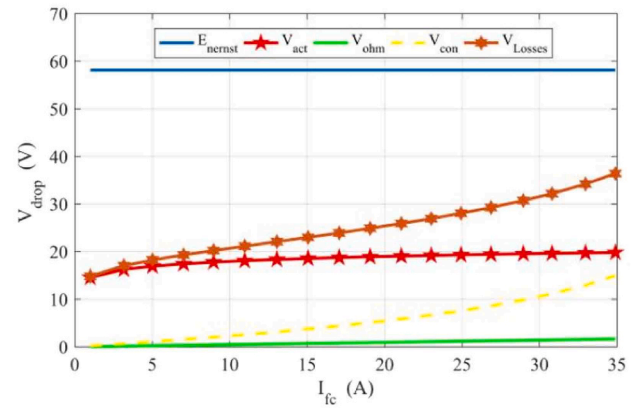
(a)



(b)



(c)



(d)

Fig. 19. Characteristics of SR-12 500 W PEMFC: (a) V/I curves, (b) P/I curves, (c) efficiency, (d) voltage losses.

3.4. Somersault foraging

The location of the food is considered a pivot in this behavior. Each individual usually swims around the pivot and somersaults to a new location. Hence, individuals change continuously their positions in relation to the best location thus far. The mathematical model for this behavior can be expressed as follows (Zhao et al., 2020):

$$x_m^n(i+1) = x_m^n(i) + 2 \cdot (rnd2 \cdot x_{best}^n - rnd3 \cdot x_m^n(i)), m = 1, \dots, k \quad (16)$$

where rnd2 and rnd3 are random numbers between (0,1).

3.5. MMRFO

The MRFO simulates the three foraging techniques of manta rays, which include chain, cyclone, and somersault foraging. The MMRFO algorithm improves the performance of the MRFO on chain foraging and cyclone foraging using the sine-cosine technique (Neggaz et al., 2024).

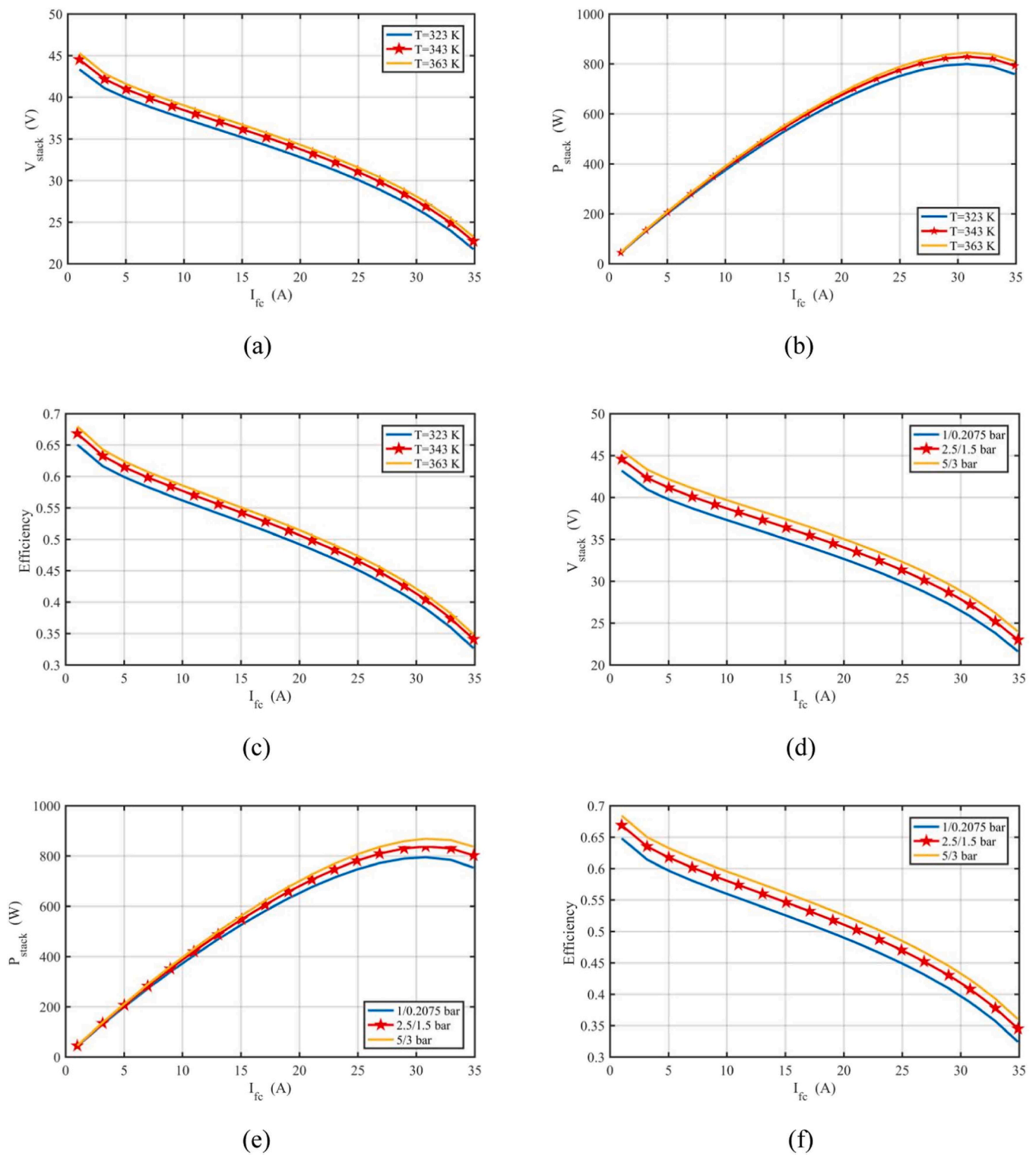


Fig. 20. Operation of SR-12 500 W PEMFC under different temperature and fuel pressure.

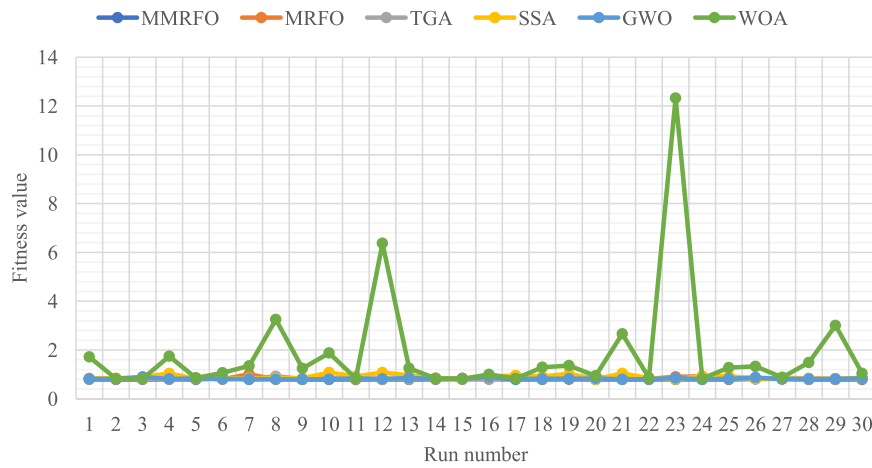


Fig. 21. The end values of the objective function (SSE) of all algorithms for Temasek 1 kW PEMFC.

The sine-cosine method has been applied to Eq. (10) and (12) to enhance the global searching capability in the exploration phase and the local searching capability in the exploitation phase. Whereas the sine-cosine function generates several solutions and fluctuates either towards or outwards of the optimal situation.

The MMRFO algorithm can switch between the chain and cyclone foraging behavior using suggested enhanced equations, which can be described as follows:

included in the study. The obtained values of the objective function in the 30 runs are recorded and statistical measurement is performed to validate the robustness and stability of the MMRFO compared with other algorithms. The search space for the unknown parameters is the same in all methods and is provided in Table 2. Furthermore, for a detailed exposition of voltage and current performance metrics for all types of PEMFCs, please see Appendix A, Tables A1 through A4.

4.1. 250 W PEMFC

$$x_{m=1}^n(i+1) = \begin{cases} x_{best}^n + rnd \cdot \sin(r4) \cdot (x_{best}^n(i) - x_m^n(i)) + B \cdot (x_{best}^n(i) - x_m^n(i))r1 < 0.5 \\ x_m^n(i) + rnd \cdot \cos(r4) \cdot (x_{best}^n(i) - x_m^n(i)) + A \cdot (x_{best}^n(i) - x_m^n(i))r1 > 0.5 \end{cases} \quad (17)$$

$$x_{m=2,\dots,k}^n(i+1) = \begin{cases} x_{best}^n + rnd \cdot \sin(r4) \cdot (x_{m-1}^n(i) - x_m^n(i)) + B \cdot (x_{best}^n(i) - x_m^n(i))r1 < 0.5 \\ x_m^n(i) + rnd \cdot \cos(r4) \cdot (x_{m-1}^n(i) - x_m^n(i)) + A \cdot (x_{best}^n(i) - x_m^n(i))r1 > 0.5 \end{cases} \quad (18)$$

$$r4 = 2 \cdot \pi \cdot rand3 \quad (19)$$

where $rand3$ and $r1$ are two random numbers in $[0,1]$. Until the stop criterion is met, all updates and computations are performed continuously. The rest of the operations of MRFO, somersault foraging, are the same. The overall process of the MMRFO is summarized in Fig. 2.

4. Simulation results and discussions

For validating the effectiveness of the MMRFO algorithm, the optimization process is performed to four commercial fuel cell stacks including, standard 250 W PEMFC, BCS-500 W PEMFC, SR-12 500 W stack, and Temasek 1 kW PEMFC stack. The electrical specifications as well as the operating variables of these fuel cell stacks are listed in Table 1. These design specifications are imported from literature. Simulation process and optimization approaches are performed under MATLAB R2018a. The optimization results obtained from the developed MMRFO algorithm are firstly compared with several methods mentioned in literature including the standard MRFO, Tree growth algorithm (TGA), salp swarm algorithm (SSA), grey wolf optimizer (GWO), whale optimization algorithm. The optimization program based on all algorithms is launched 30 individual times for all PEMFC stacks

The 250 W PEMFC is utilized as a standard benchmark for evaluating the developed approach. For better validation of the effectiveness of the suggested methodology, the attainments are compared with MRFO, TGA, SSA, GWO, and WOA. The values of the OF for all algorithms over the 30 individual runs on the optimization program are shown in Fig. 3. The end values of the convergence curve over the 30 runs are listed in Table 3. Also, Statistical study based on the end values of convergence curve over the 30 individual runs is presented in Table 3. The table is organized based on various metrics including the best value of the objective function obtained so far (best), the maximum value of the objective function (worst), the relative error (RE), the standard deviation (SD), the root mean square error (RMSE), and the efficiency which is considered the percentage ratio between the minimum value of the objective function obtained so far to the average value (mean) of the objective function over the 30 runs. Based on Fig. 3, it can be observed that the value of the objective function from MMRFO (blue line) lies within a narrow range which is translated in the lower values of the statistical metrics such as median, RE, SD, RMSE. As a result, the value of the efficiency of the MMRFO is the highest compared to others. A radar plot for the best values of the objective function (SSE) for the studied methods is shown in Fig. 4.

The obtained results, visualized through a box plot in Fig. 5, underscore the stability of the MMRFO algorithm. This stability is not

Table 9
Comparison of the achievements of MMRFO with other algorithms for Temasek 1 kW PEMFC.

Algorithm	MMRFO	MRFO	TGA	SSA	GWO	WOA
1	0.796745313	0.815349061	0.809462168	0.83731553	0.814537086	1.714717926
2	0.792375705	0.800757942	0.824319655	0.838868634	0.811746264	0.822550764
3	0.791930545	0.843457278	0.808457028	0.891246966	0.892941468	0.825093268
4	0.799345324	0.820706442	0.808645786	1.039184416	0.816077822	1.745325092
5	0.79098801	0.849089859	0.852346624	0.819773013	0.832337392	0.853077596
6	0.848622435	0.802903712	0.814864832	0.819167095	0.803694266	1.065172121
7	0.792224601	1.008512002	0.817928547	0.84013104	0.813977731	1.348375838
8	0.816960065	0.79681998	0.922162916	0.839170739	0.79661968	3.25141167
9	0.792428256	0.799807817	0.813655709	0.842391075	0.794618388	1.245742187
10	0.792539647	0.823209514	0.836871157	1.075200377	0.803290845	1.87693034
11	0.791870399	0.799372164	0.83339572	0.921400968	0.837355073	0.83443356
12	0.792242625	0.81765284	0.796926618	1.0763793	0.814297625	6.375099889
13	0.791096108	0.806283212	0.813503405	0.962598274	0.85469453	1.252712348
14	0.798412893	0.808400058	0.836152341	0.841813303	0.820053769	0.835996755
15	0.837330085	0.793815717	0.809173531	0.824357033	0.818360314	0.81620668
16	0.80596794	0.805653427	0.813010309	0.910616829	0.896205629	1.000420387
17	0.792073179	0.937726541	0.860332483	0.950828653	0.816587903	0.828402727
18	0.793988342	0.86834442	0.811970211	0.9101652	0.820889256	1.293083826
19	0.80084017	0.826607322	0.853136902	1.02818525	0.824592859	1.361069897
20	0.793499528	0.805453571	0.924600759	0.796814717	0.834591193	0.943503244
21	0.792250858	0.854703402	0.797756182	1.043596877	0.799939305	2.662858256
22	0.792978833	0.818665797	0.815396087	0.841865339	0.80506791	0.859878726
23	0.792023044	0.902130207	0.805761774	0.807777765	0.851384035	12.32351569
24	0.792781377	0.92423439	0.798995403	0.929354218	0.798956632	0.835990654
25	0.791997356	0.818388123	0.823522931	0.930625278	0.821276852	1.277521315
26	0.815679726	0.830086137	0.807283302	0.827472895	0.87380016	1.32942144
27	0.818294507	0.821919361	0.85881839	0.817438108	0.802063583	0.878447658
28	0.792309385	0.810287684	0.836623683	0.817717349	0.804877838	1.487296147
29	0.79212204	0.797381514	0.819909781	0.823083332	0.81062557	3.008021033
30	0.792443173	0.806874375	0.858274828	0.828967947	0.837773826	1.044361288
Min	0.79098801	0.793815717	0.796926618	0.796814717	0.794618388	0.81620668
Max	0.848622435	1.008512002	0.924600759	1.0763793	0.896205629	12.32351569
Mean	0.799478716	0.833819796	0.829441969	0.891116917	0.824107827	1.866554611
Median	0.79249141	0.818020482	0.816662317	0.841839321	0.816332862	1.249227267
SD	1.421473229	4.935560361	3.165014838	8.662281521	2.648961786	227.0078515
RE	0.322029111	1.511840017	1.224028021	3.550469077	1.113343424	38.60595449
MAE	0.008490706	0.040004079	0.03251535	0.0943022	0.029489439	1.050347931
RMES	0.016352841	0.062889608	0.045006542	0.127068092	0.039343829	2.466720906
Eff.	98.96709349	95.49216135	96.20719293	90.17639615	96.51402612	67.69257376

merely a result of algorithmic robustness but also reflects a comprehensive statistical validation where the algorithm consistently outperformed its counterparts across multiple runs. Specifically, the run yielding the optimal value of the objective function, highlighted in Fig. 6, showcases the superior convergence characteristics of MMRFO, a pivotal factor in its effectiveness. This assertion is supported by quantitative metrics, with the MMRFO technique achieving the lowest SSE at 0.642089, as detailed in Table 3. To further substantiate these findings, we conducted a rigorous comparative analysis against contemporary algorithms. The convergence curves, depicted in Fig. 6, reveal not only the rapidity with which MMRFO approaches the optimal solution but also its computational efficiency, making it particularly suitable for complex optimization problems like those encountered in FC design. This computational efficiency is mirrored in the real-world applicability of our findings, where the optimized parameters derived from MMRFO, detailed in Table 4, were integrated into a simulation model for a 250 W PEMFC.

The generated polarization curves based on the identified parameters are compared with those given in the data sheet of the producer. As can be observed in Fig. 7.a and 7.b, the MMRFO gives better agreement between the measured and calculated polarization characteristics which is translated in the best value of SSE. The fuel cell efficiency mentioned in Fig. 7.c denotes the voltage efficiency. Moreover, the nuanced analysis of voltage loss variations within the fuel cell stack, presented in

Fig. 7.d, offers invaluable insights into the loss mechanisms at play. By dissecting these losses in relation to cell current, our study not only elevates the understanding of fuel cell operation but also highlights the potential of MMRFO in mitigating inefficiencies, thereby pushing the boundaries of fuel cell technology towards more sustainable and efficient energy solutions.

The dynamic performance of the 250 W PEMFC under various operating temperatures and different pressures of the input gases has been also investigated. In the case of changing the cell temperature the pressure of the input gases is maintained constant at the values reported in the data sheet of the manufacturer. Similarly, while changing the pressure on the reactants the cell temperature is kept constant at the value mentioned in Table 1. The characteristics of the 250 W PEMFC stack under different operating conditions are presented in Fig. 8.

4.2. BCS 500 W PEMFC

In this phase of our investigation, the BCS 500 W PEMFC was selected as the second benchmark fuel cell stack for comprehensive analysis. The main design and operating factors of the fuel cell are listed in Table 2. The values of the objective function for MMRFO and other algorithms over the 30 individual runs of the optimization program are shown in Fig. 9. A statistical study based on the end values of the objective function over the 30 individual runs is presented in Table 5.

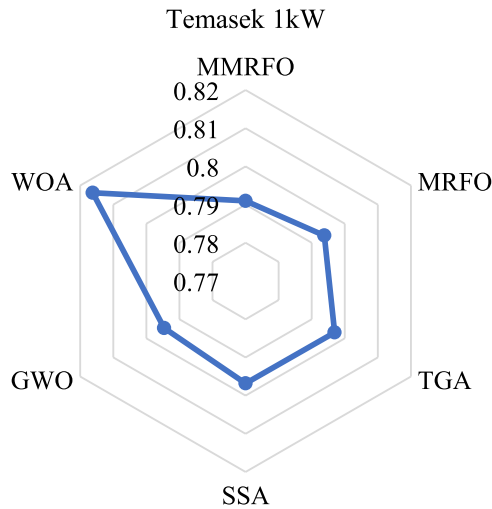


Fig. 22. Radar plot for the best OF obtained so far from all algorithms for Temasek 1 kW PEMFC.

Based on Fig. 9, it is observed that the value of the objective function (SSE) from MMRFO (blue line) varies within a narrow domain except for three overshoots that ensures the superior values of RE, SD, and RMSE compared to other algorithms. A radar plot for the best values of the objective function (SSE) for the studied techniques is shown in Fig. 10.

Fig. 11 displays a box plot of the obtained results, corroborating the stability of the MMRFO methodology. It captures the optimal values of the objective function across all considered algorithms. Furthermore, Fig. 12 illustrates the convergence curves associated with these optimal runs, providing insight into the algorithms' performance over iterations. As provided in Fig. 12, the MMRFO algorithm shows the best convergence characteristics. As observed from Fig. 12 and Table 5 the MMRFO technique has the best effectiveness compared with the other methods with the best SSE of 0.011574603. The values of the identified parameters based on all methods are given in Table 6. The values of the optimized parameters from the MMRFO are fed to the simulation model to generate the polarization curves of the BCS 500 W PEMFC. The generated polarization curves based on the identified parameters are compared with those given in the data sheet of the producer. As can be observed in Fig. 13.a, 7.b, and 13.c, the MMRFO gives better agreement between the measured and calculated polarization characteristics which is translated in the best value of SSE. In addition, the variation of different types of over-voltages occurred in the fuel cell stack with respect to the cell current is provided in Fig. 13.d. The characteristics of the BCS 500 W PEMFC stack under various operating conditions are presented in Fig. 14.

4.3. SR-12 500 W PEMFC

The SR-12 500 W PEMFC is a standard benchmark fuel cell stack studied that is commonly employed for evaluation of various parameters' identification approaches. The values of the objective function for

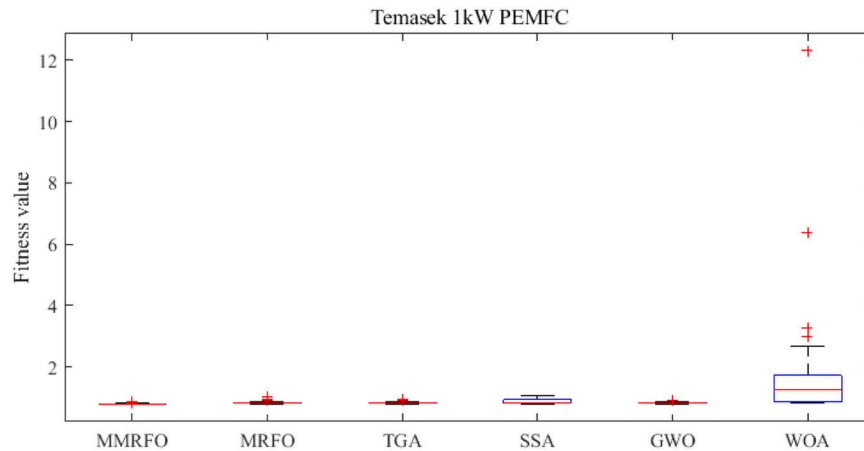


Fig. 23. Box-plot for Temasek 1 kW PEMFC.

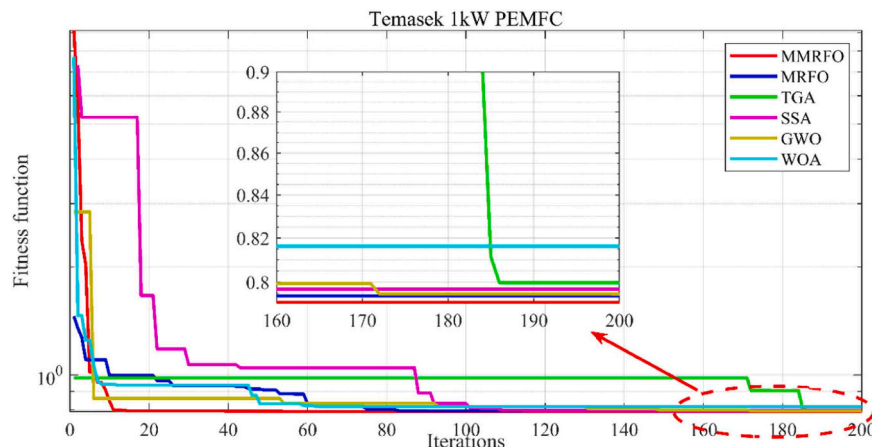
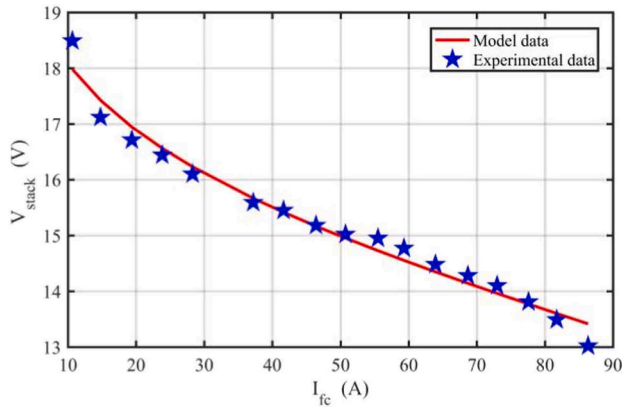


Fig. 24. Convergences curves of the best run for all algorithms.

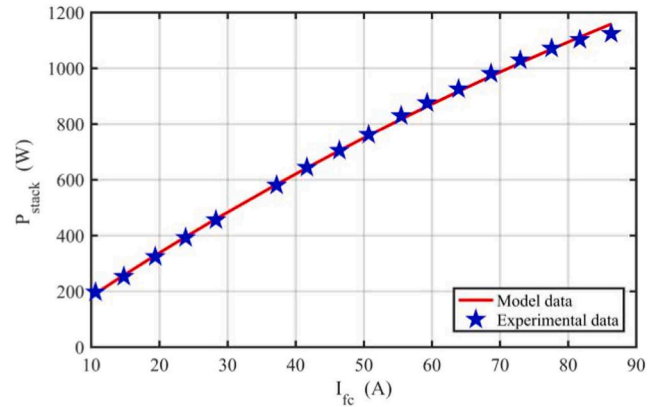
Table 10

Extracted parameters of Temasek 1 kW PEMFC based on the MMRFO and other methods.

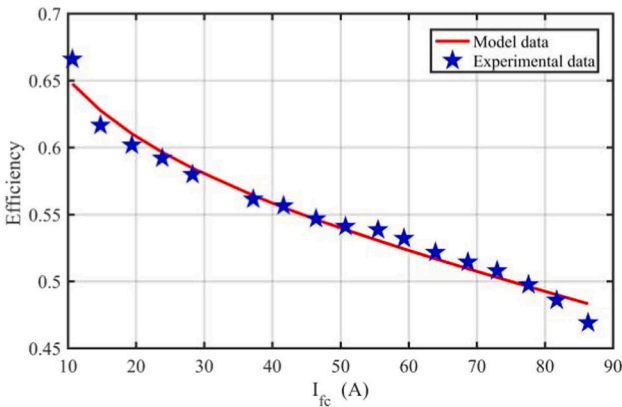
	ξ_1	$\xi_2 \times 10^{-3}$	$\xi_3 \times 10^{-5}$	$\xi_4 \times 10^{-4}$	λ	β	$R_c \times 10^{-4}$	SSE
MMRFO	−0.98118	3.72823	6.421681	−2.30425	13.01261	0.066827	1.00015	0.79099
MRFO	−1.176938	4.57888	8.81969	−1.72465	17.79465	0.12465	7.54113	0.79382
TGA	−0.87218	2.526	3.818	−2.423	14.792	0.665	0.894	0.79692
SSA	−1.1336	4.1130	0.6004	−2.231	21.5365	0.1025	0.0001	0.796814
GWO	−1.1693	3.8318	0.3690	−2.2060	17.0946	0.1030	0.0001	0.795101
WOA	−1.0558	4.2307	0.7605	−2.6000	16.8066	0.0370	0.0001	0.826329
ICEFO (Sultan et al., 2021b)	−1.07372	4.2503	7.8165	−2.3196	13.00342	0.06370	1.0109	0.792421
ISSA (Sultan et al., 2020)	−1.1821	4.582	9.630	−2.286	18.386	9.60	1.03	0.79268
CHHO4 (Menesy et al., 2019)	−1.094374	4.428226	8.765584	−2.14649	18.63921	0.10155	1.89107	0.802341
SBO (Duan et al., 2019)	−1.0312	2.409	3.950	−0.953	9.985	0.1269	1.123	1.63220
SSO (Rao et al., 2019)	−1.0299	2.4105	1.00	−0.9540	10.0005	0.1274	1.0873	1.6481



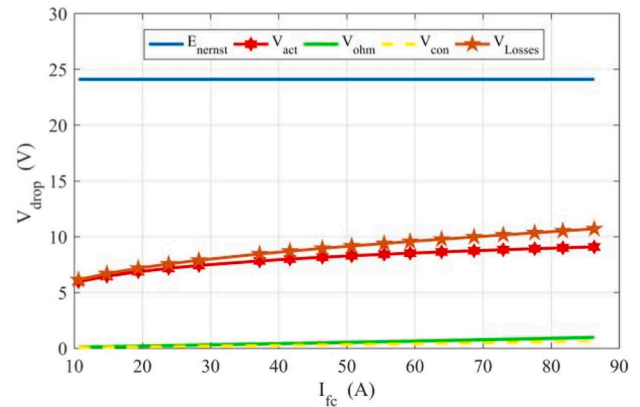
(a)



(b)



(c)

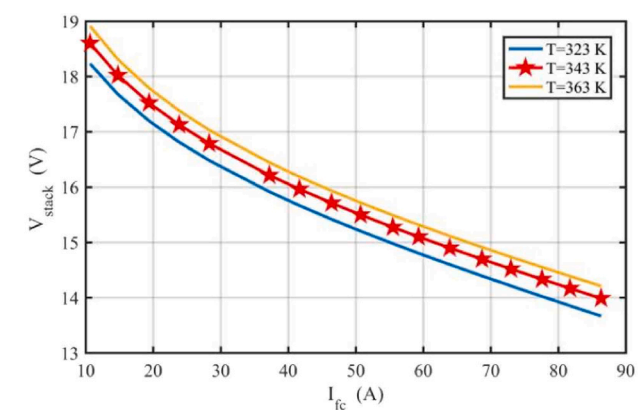


(d)

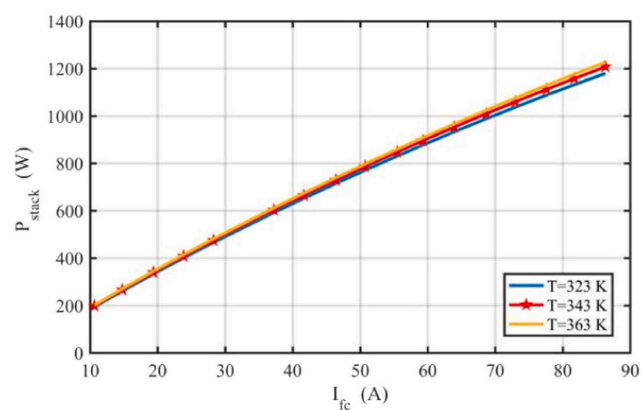
Fig. 25. Characteristics of Temasek 1 kW PEMFC: (a) V/I curves, (b) P/I curves, (c) efficiency, (d) voltage losses.

MMRFO and other algorithms over the 30 individual runs of the optimization program are shown in Fig. 15. A statistical study based on the end values of the objective function over the 30 individual runs is presented in Table 7. Based on Fig. 15, the reader can notice that the objective function (SSE) from MMRFO (blue line) has the lowest observations for the 30 runs and it is reflected on the best results of the statistical study provided in Table 7. A radar plot for the best values of the SSE for all techniques is shown in Fig. 16. A box plot for the obtained results is shown in Fig. 17. The box plot confirms the stability of the MMRFO algorithm. The convergence trends for the MMRFO and other algorithms are depicted in Fig. 18. As shown from Fig. 18, the MMRFO algorithm gives the best convergence characteristics. As observed from

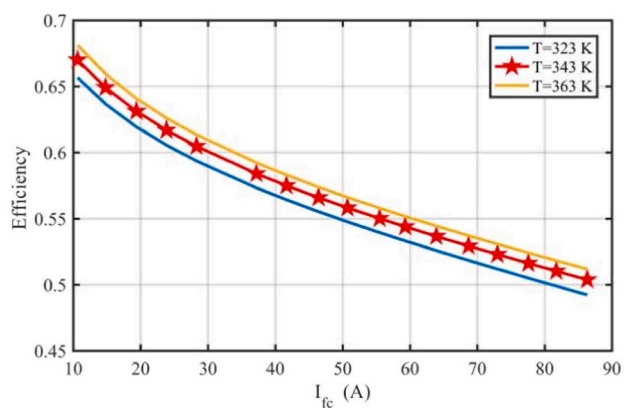
Fig. 17 and Table 7 the MMRFO technique is a robust and reliable method for identifying the parameters of SR-12 500 W PEMFC with the lowest value of the SSE of 1.056651541. The values of the identified parameters based on all methods are given in Table 8. The values of the optimized parameters from the MMRFO are fed to the simulation model to generate the polarization curves of the SR-12 500 W PEMFC. The generated polarization curves based on the identified parameters are compared with those given in the data sheet of the producer. As can be observed in Fig. 19.a, 19.b, and 19.c, the MMRFO gives better agreement between the measured and calculated polarization characteristics which is translated in the best value of SSE. In addition, the variation of different types of voltage loss occurred in the fuel cell stack with respect



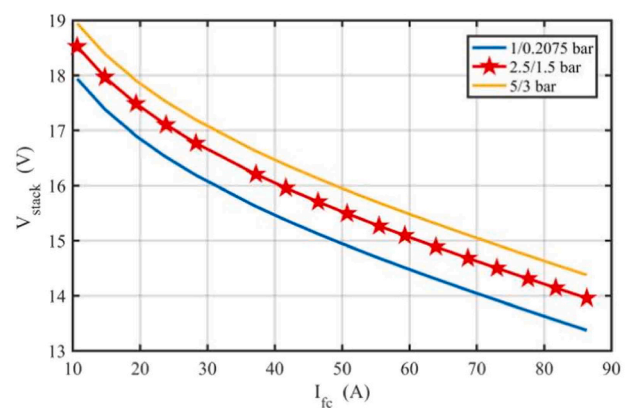
(a)



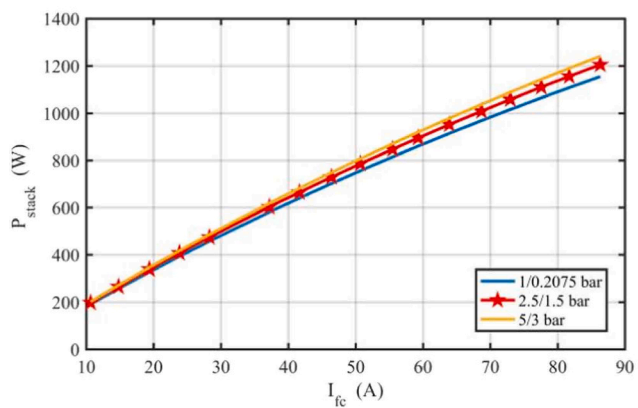
(b)



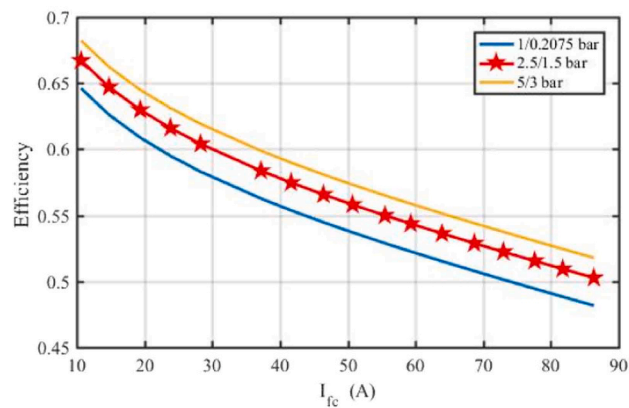
(c)



(d)



(e)



(f)

Fig. 26. Operation of Temasek 1 kW PEMFC under different temperature and fuel pressure.

Table 11
Results of sensitivity analysis for 250 W PEMFC.

variable	Percentage change (%)	250 W PEMFC	
		Objective function	Y (%)
Original case	0 %	0.642082942003649	2.29120487175478
ξ_1	+10 %	119.362827407796	19.0451422115348
	+5 %	36.3285137044558	10.4912907347379
	−5 %	17.0176711468008	6.61641221886309
ξ_2	−10 %	80.7411422925669	15.1702636956671
	+10 %	89.1124337080471	15.9619629046647
	+5 %	18.8870803311161	7.01226182336169
ξ_3	−5 %	39.0915775669841	10.8871403392365
	−10 %	129.521428179699	19.8368414205324
	+10 %	21.6382971930548	8.05667375762715
ξ_4	+5 %	8.79650873964561	4.99705650778402
	−5 %	1.88915582646347	1.78980843219893
	−10 %	7.82359136671918	4.18179524175938
λ	+10 %	6.21200221921488	4.04606822151388
	+5 %	3.87741170960450	3.15430321472942
	−5 %	1.22015194273682	1.63430125773449
β	−10 %	0.897482685487883	1.39871138870665
	+10 %	1.24266516883126	1.61828304643872
	+5 %	1.64415272468872	1.89938293676873
R_c	−5 %	3.00617401716767	2.73193054827350
	−10 %	4.09960117881330	3.23133605563056
	+10 %	4.82311203343507	3.50872011938352
	+5 %	3.34093098692853	2.89996249557053
	−5 %	1.44070392498310	1.72524274424814
	−10 %	1.02265790954887	1.43238450606702
	+10 %	2.28710911096472	2.33663703051365
	+5 %	2.25007407088430	2.31392095113559
	−5 %	2.17727175064321	2.26848879237397
	−10 %	2.14150447048273	2.24577271299041

Table 12
Results of sensitivity analysis for BCS 500 W PEMFC.

variable	Percentage change (%)	BCS 500 W PEMFC	
		Objective function	Y (%)
Original case	0 %	0.0115746028716271	0.0618716395215303
ξ_1	+10 %	486.975751715879	24.4854393907314
	+5 %	204.921838102973	15.8832442750226
	−5 %	1.42879129003698	1.32114595639507
ξ_2	−10 %	79.9896580900063	9.92334107210387
	+10 %	58.5718645883516	8.49132755625011
	+5 %	0.308796875192064	0.605139198468192
ξ_3	−5 %	186.864027845852	15.1672375170957
	−10 %	431.682326529672	23.0534258748776
	+10 %	106.033015533351	11.4248897529986
ξ_4	+5 %	71.0661946872694	9.35296945615620
	−5 %	22.0529853546199	5.20912886247131
	−10 %	8.00659686805191	3.13720856562888
λ	+10 %	75.4611939427965	9.70761606221942
	+5 %	57.9929036888941	8.49433261076658
	−5 %	30.7010367591912	6.06776570786092
β	−10 %	20.8774600833906	4.85448225640809
	+10 %	38.5149999914282	6.84771707954298
	+5 %	40.6314640277404	7.05323829417466
R_c	−5 %	45.9125904156100	7.53498487344712
	−10 %	49.2465574625739	7.81981434247807
	+10 %	46.1375262689112	7.56048425902312
	+5 %	44.5805111493101	7.42076670916844
	−5 %	41.6145467040583	7.14133160945907
	−10 %	40.2055973784075	7.00161405960439
	+10 %	43.3858931252036	7.30949140213396
	+5 %	43.2291800868040	7.29527028072386
	−5 %	42.9169067478691	7.26682803790365
	−10 %	42.7613464473338	7.25260691649355

Table 13
Results of sensitivity analysis for SR-12 500 W PEMFC.

variable	Percentage change (%)	SR-12 500 W PEMFC	
		Objective function	Y (%)
Original case	0 %	1.05665154120048	0.612952378343352
ξ_1	+10 %	594.261320738603	17.8689560112362
	+5 %	150.495565524248	8.96743810773526
	−5 %	145.982228056790	8.83559769926664
ξ_2	−10 %	585.234645803684	17.7371156027676
	+10 %	626.786607843442	18.3570142712884
	+5 %	156.330930156637	9.14554703352705
ξ_3	−5 %	161.001421264463	9.27738744199568
	−10 %	636.127590059093	18.4888546797570
	+10 %	51.5857925575842	5.21333400642989
ξ_4	+5 %	14.0245863076477	2.63962710533210
	−5 %	12.7196375255709	2.50778669686348
	−10 %	48.9758949934305	5.08149359796127
λ	+10 %	4.45818845485732	1.37528294441541
	+5 %	2.01327648364588	0.812640808837274
	−5 %	1.62596289484947	0.814271049705124
β	−10 %	3.68356127726446	1.27902058758464
	+10 %	1.05684498993831	0.608703446868567
	+5 %	1.05877174651396	0.610715100120874
R_c	−5 %	1.09115652557174	0.618822359657058
	−10 %	1.12881349394424	0.629130580375691
	+10 %	10.7665233470285	2.127440588626231
	+5 %	3.65608445864510	1.11865553457335
	−5 %	3.00587386202292	1.10000287639473
	−10 %	9.46610215378406	1.98759540538989
	+10 %	1.21328542204261	0.647856795105541
	+5 %	1.12083839032132	0.627578136017983
	−5 %	1.05837414200843	0.608317995343886
	−10 %	1.08835692541682	0.611709177236067

Table 14
Results of sensitivity analysis for Temasek 1 kW PEMFC.

variable	Percentage change (%)	Temasek 1 W PEMFC	
		Objective function	Y (%)
Original case	0 %	0.790988010229386	1.08476192338759
ξ_1	+10 %	70.1148160410498	12.9198800112947
	+5 %	18.1239851531346	6.45136953699988
	−5 %	18.1158246123341	6.48565141158971
ξ_2	−10 %	70.0984949594487	12.9541618858845
	+10 %	105.191539661390	15.8949583726342
	+5 %	26.8886220317154	7.95604965496451
ξ_3	−5 %	26.8986375969310	7.92176778037471
	−10 %	105.211570791820	15.8606764980443
	+10 %	9.36312451514498	4.53159402305017
ξ_4	+5 %	2.93473960167038	2.30070998592774
	−5 %	2.93186974082198	2.38914421020131
	−10 %	9.35738479344811	4.56587589763999
λ	+10 %	6.46046739112441	3.68341412028686
	+5 %	2.20906453102278	1.96685389719163
	−5 %	2.20623782874426	2.06660639842709
β	−10 %	6.45481398656737	3.81112531615853
	+10 %	0.837040791790537	1.05625914209289
	+5 %	0.804099567950438	1.04965143499819
R_c	−5 %	0.806204510708869	1.13081664667909
	−10 %	0.861895398509093	1.18292576033999
	+10 %	0.816829740723825	1.14392486350867
	+5 %	0.797452463493045	1.11434339344793
	−5 %	0.797436380932841	1.05626135220516
	−10 %	0.816797575603407	1.05209111221983
	+10 %	0.793710044822709	1.10186301982158
	+5 %	0.791832841181294	1.09331247160438
	−5 %	0.791175551966992	1.07621137517001
	−10 %	0.792395466394095	1.06766082695282

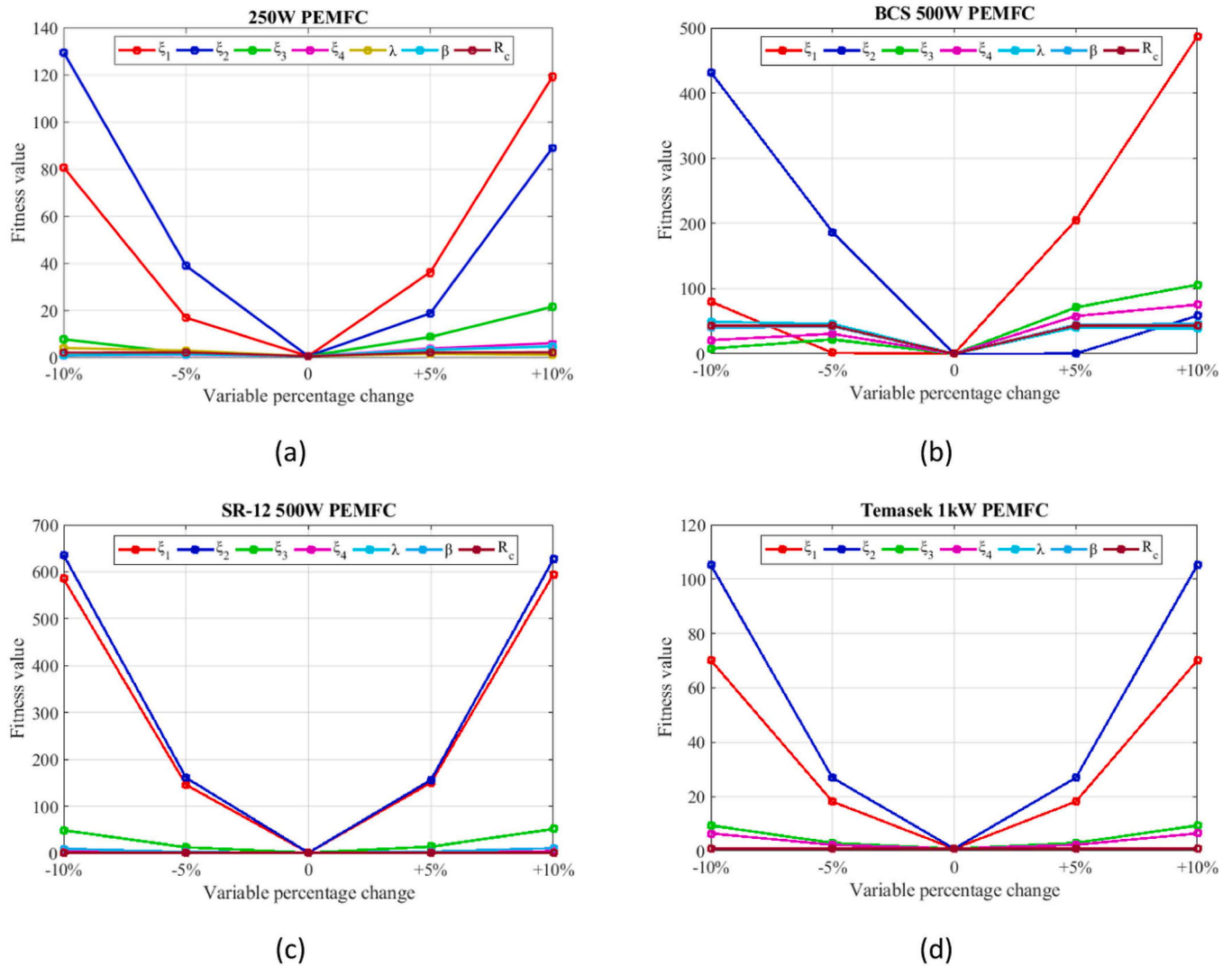


Fig. 27. Impact of percentage change in design variables on the objective function.

to the cell current is provided in Fig. 19.d. The operation of the SR-12 500 W PEMFC stack under different scenarios are presented in Fig. 20.

4.4. Temasek 1 kW PEMFC

The fourth fuel cell stack used for validating the identification method based on the MMRFO is the Temasek 1 kW stack. Fig. 21 shows the values of the objective function for MMRFO and other algorithms over the 30 individual runs. Statistical results are presented in Table 9. Based on Fig. 21 and Table 9, the reader can notice that the MMRFO has the best results compared with other methods. A radar plot for the best values of the objective function for all techniques is shown in Fig. 22. A box plot for the obtained results is shown in Fig. 23. The box plot confirms the stability of the MMRFO algorithm. The convergence trends for the MMRFO and other algorithms are shown in Fig. 24. The values of the identified parameters based on all algorithms are provided in Table 10. The polarization curves based on the identified parameters for the Temasek 1 kW stack are compared with that given in the data sheet of the producer and given in Fig. 25. As can be observed in Fig. 25.a, 25.b and 25.c, the MMRFO gives better agreement between the measured and calculated polarization characteristics which is translated in the best value of SSE. In addition, the variation of different types of voltage loss occurred in the fuel cell stack with respect to the cell current is provided in Fig. 25.d. The characteristics of the Temasek 1 kW PEMFC stack under

different operating conditions are presented in Fig. 26.

4.5. Sensitivity analysis

In this section sensitivity analysis has been provided, in which the impact of changing the design variables on the fitness value has been investigated. To measure the sensitivity a function (Y) has been defined.

$$Y = \frac{1}{N} \sum_{m=1}^M \left| \frac{V_{cal}(m) - V_{meas}(m)}{V_{meas}(m)} \right| \times 100$$

The percentage change of each parameter varied from $\pm 10\%$. The obtained results of the objective function as well as the function “Y” in the four case studies are provided in Tables 11–14. From the obtained results it is shown that the first two variables ξ_1 and ξ_2 has the significant impact on the objective function. The obtained results are graphically presented in Figs. 27 and 28.

5. Conclusion

This paper has proposed an effective approach for optimal identification of PEMFC parameters. The developed approach is based on employing the modified version of Manta Ray Foraging Optimization (MMRFO) for minimizing the sum of squared errors (SSE) between the measured data and the optimally attained mathematical model. The

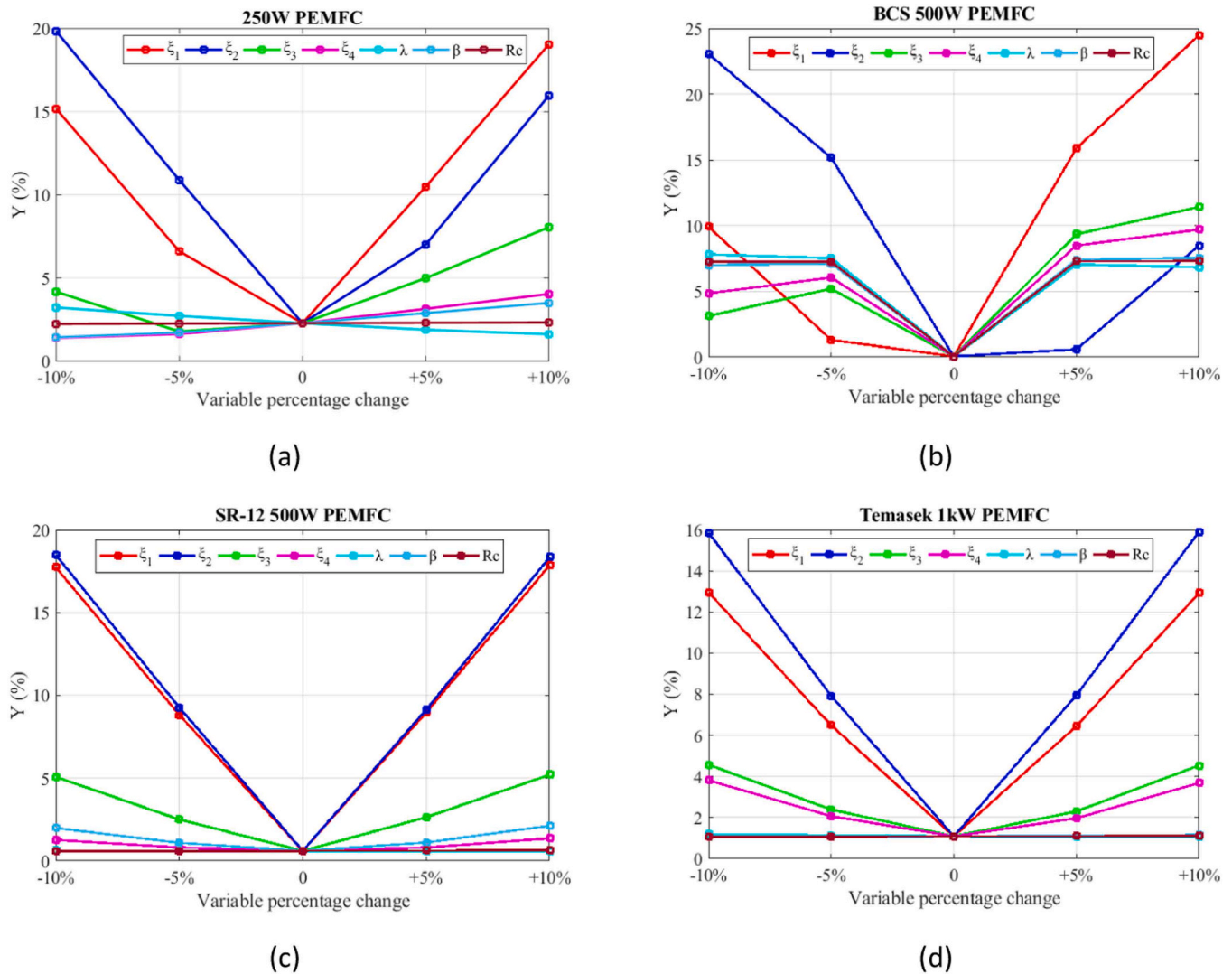


Fig. 28. Impact of percentage change in design variables on the function “Y”.

system analysis is based on MMRFO-based PEMFC model comprising design specifications from 250 W PEMFC, BCS-500 W PEMFC, SR-12 500 W PEMFC, and 1 kW Temasek PEMFC stacks. The simulations have been accomplished in MATLAB environment and the attained results of the MMRFO were compared with some recent metaheuristic techniques including TGA, SSA, GWO, WOA, and MRFO. For future work, the MMRFO algorithm used in this study could be expanded to address other complex optimization problems beyond PEMFC parameters estimation. Additionally, exploring the dynamic performance of PEMFCs using the MMRFO algorithm for enhanced operational insights, as well as applying this methodology to different fuel cell types like Solid Oxide Fuel Cells (SOFCs).

CRedit authorship contribution statement

Hamdy M. Sultan: Writing – original draft, Supervision, Software,

Methodology, Investigation, Conceptualization. **Ahmed S. Menesy:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ahmed Korashy:** Writing – review & editing, Writing – original draft, Software, Resources, Methodology. **Abdelazim Hussien:** Writing – original draft, Visualization, Validation, Software, Resources, Formal analysis. **Salah Kamel:** Visualization, Validation, Resources, Project administration, Methodology.

Declaration of Competing Interest

Authors has no conflict of interest.

Data Availability

Data will be made available on request.

Appendix A

Table A1

Comprehensive dataset of voltage and current measurements for BCS-500 W PEMFC stack.

No.	I_{fc} (A)	$V_{stack, meas}$ (V)
1	0.600	29.0000
2	2.100	26.3100
3	3.580	25.0900
4	5.080	24.2500
5	7.170	23.3700
6	9.550	22.5700
7	11.390	22.0600
8	12.540	21.7500
9	13.730	21.4500
10	15.730	21.0900
11	17.020	20.6800
12	19.110	20.2200
13	21.200	19.7600
14	23	19.3600
15	25.080	18.8600
16	27.170	18.2700
17	28.060	17.9500
18	29.260	17.3000

Table A2

Comprehensive dataset of voltage and current measurements for 250 W PEMFC stack.

No.	I_{fc} (A)	$V_{stack, meas}$ (V)
1	0.4717	21.63
2	2.149	19.68
3	2.83	18.78
4	3.983	17.88
5	5.713	17.58
6	7.075	17.12
7	8.019	16.77
8	11.16	15.92
9	13.73	15.22
10	16.56	14.42
11	17.56	14.02
12	19.03	13.52
13	20.34	12.82
14	22.01	10.86
15	22.96	9.058

Table A3

Comprehensive dataset of voltage and current measurements for SR-12 500 W PEMFC stack.

No.	I_{fc} (A)	$V_{stack, meas}$ (V)
1	1.004	43.170
2	3.166	41.140
3	5.019	40.090
4	7.027	39.040
5	8.958	37.99
6	10.97	37.08
7	13.05	36.03
8	15.06	35.19
9	17.07	34.07
10	19.07	33.02
11	21.08	32.04
12	23.01	31.20
13	24.94	29.80
14	26.87	28.96
15	28.96	28.12
16	30.81	26.30
17	32.97	24.06
18	34.90	21.40

Table A4
Comprehensive dataset of voltage and current measurements for Temasek 1 kW PEMFC stack.

No.	I _{fc} (A)	V _{stack,meas} (V)
1	10.64	18.49
2	14.77	17.12
3	19.38	16.71
4	23.84	16.44
5	28.29	16.1
6	28.29	16.1
7	37.2	15.59
8	41.65	15.45
9	46.42	15.18
10	50.72	15.02
11	55.49	14.95
12	59.3	14.77
13	63.92	14.48
14	68.69	14.28
15	72.98	14.1
16	77.59	13.81
17	81.73	13.49
18	86.34	13.02

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